Unit-II: Physical Layer and Data Link Layer

Physical Layer:

- **Data and Signals in Physical layer**

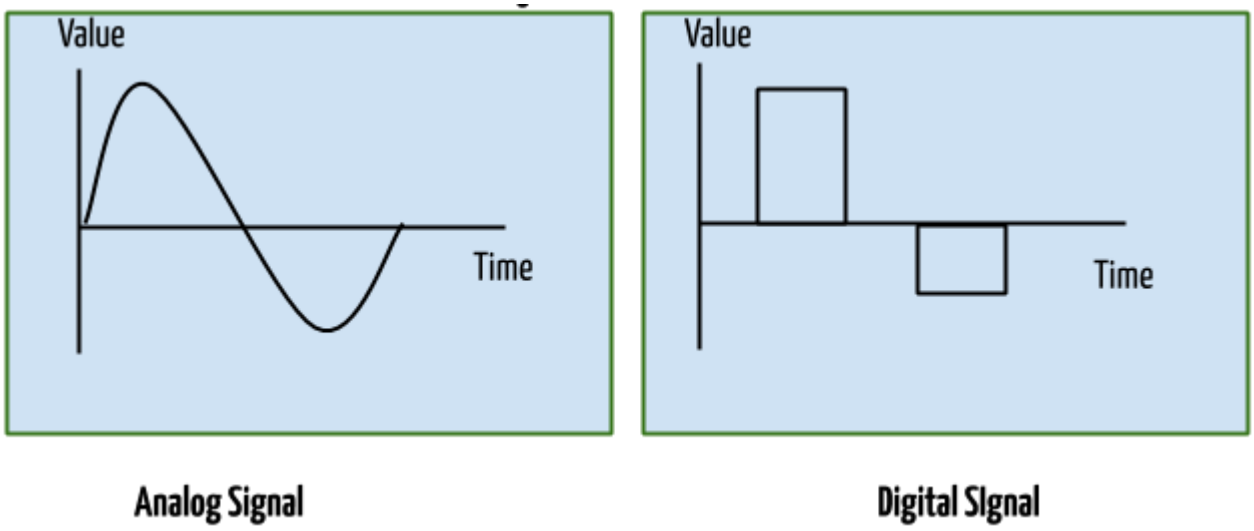
One of the major role of **Physical layer** is to transfer the **data in form of signals** through a transmission medium. It doesn't matter what data you are sending, it can be text, audio, image, video etc. everything is transferred in form of signals. This happens because a data cannot be send as it is over a transmission medium, it must be converted to a form that is acceptable by the transmission media, signals are what a transmission medium carry. In this guide, we will discuss data and signals.

Analog and Digital

Both the data and the signal can be represented in form of analog and digital.

Analog and Digital Data:
Analog data is continuous data that keeps changing over time, for example in an analog watch, the hour, minute and second hands keep moving so you infer the time by looking at it, it keeps changing. On the other hand digital watch shows you discrete data such as 12:20 AM, 5:30 PM etc. at a particular moment of time.

Analog and Digital Signals:
Similar to data, a signal can be analog or digital. An analog signal can have infinite number of values in a given range, on the other hand a digital signal has limited number of values in a given range. The following digram shows analog and digital signals.



1. Analog Signals

An analog signal can be be categorized as **simple** or **composite**.

1.1 Simple Analog signal

A simple analog signal can be represented in form of sine wave. A sine wave is shown in the above diagram.

A simple analog signal is smooth, consistent and continuous. As you can see in the diagram above that a arc above the time axis is followed by the similar arc below the time axis and so on.

Parameters of simple Analog signal – Sine wave

There are three parameters that defines a sine wave – peak amplitude, frequency and phase.

Peak amplitude: Absolute value of highest intensity of sine wave.

Frequency and Period: Period is the amount of time a signal takes to complete one cycle, it is denoted by T. Frequency refers to the number of cycles in 1 second, it is denoted by f. They are inversely proportional to each other which means f = 1/T.

Phase: Phase refers to the position of sine wave relative to the time 0. For example if the sine wave is at its highest intensity at the time zero then the phase value for this sine wave is 90 degrees. Phase is measured in degrees or radians.

1.2 Composite Analog signal

Unlike sine wave which is smooth and consistent, composite analog signals or waves are not smooth and consistent, which means an arc above the time axis doesn't necessarily followed by arc below the time axis. You can imagine them as a group of sine waves with different frequency, amplitude and period.

Bandwidth: The range of frequencies in a composite signal is called bandwidth. For example if a composite signal contains waves with the frequencies ranging from 2000 to 4000 then you can say that the bandwidth of this composite signal is 4000-2000 = 2000Hz. Bandwidth is measured in Hz.

2. Digital Signals

Similar to analog signals, data can be transmitted in form of digital signals. For example a data that is converted it into a machine language (combination of 0s and 1s) such as 1001 can be represented in form digital signals. 1 represents high voltage and 0 represents low voltage.

Bit Rate: A bit rate is measured as bits per second, it represents the number of 1s send in 1 second.

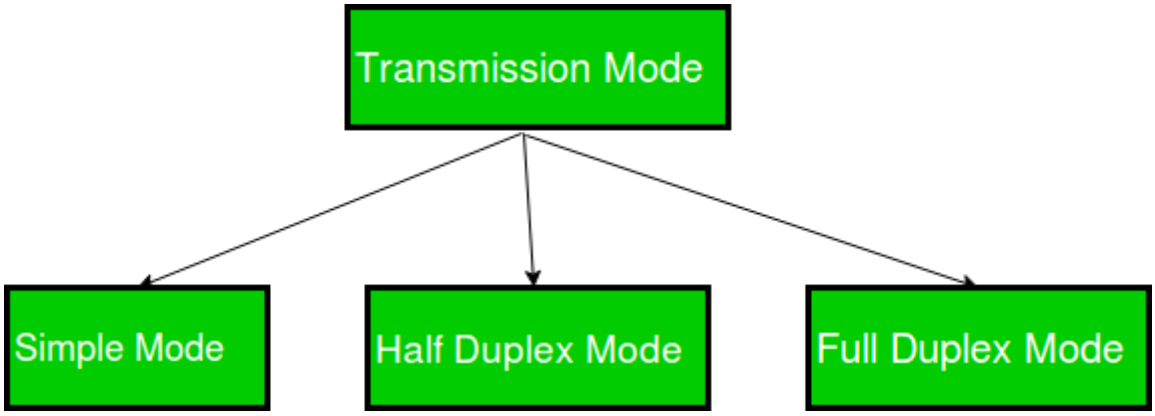
Bit Length: A bit length is the distance a bit occupies on the transmission medium.

Now that you are familiar with analog and digital signals, it's time time to cover digital transmission which we will discuss in the next chapter.

❖ Transmission Modes in Computer Networks (Simplex, Half-Duplex and Full-Duplex)

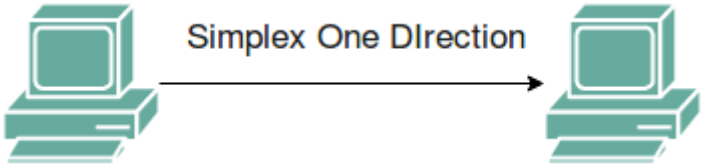
Transmission mode means transferring data between two devices. It is also known as a communication mode. Buses and networks are designed to allow communication to occur between individual devices that are interconnected.

There are three types of transmission mode:-



These are explained as following below.

1. Simplex Mode –
In Simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit, the other can only receive. The simplex mode can use the entire capacity of the channel to send data in one direction.
Example: Keyboard and traditional monitors. The keyboard can only introduce input, the monitor can only give the output.

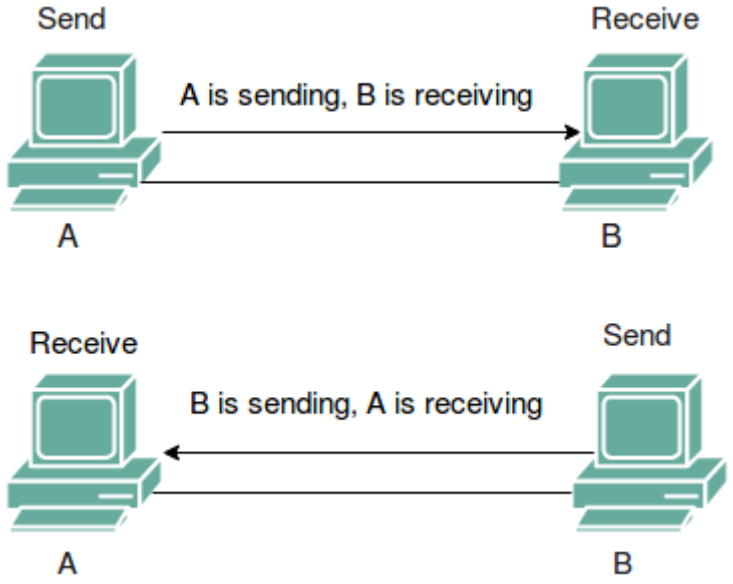


Advantages:
- Simplex mode is the easiest and most reliable mode of communication.
- It is the most cost-effective mode, as it only requires one communication channel.
- There is no need for coordination between the transmitting and receiving devices, which simplifies the communication process.
- Simplex mode is particularly useful in situations where feedback or response is not required, such as broadcasting or surveillance.

Disadvantages:
- Only one-way communication is possible.
- There is no way to verify if the transmitted data has been received correctly.
- Simplex mode is not suitable for applications that require bidirectional communication.

2. Half-Duplex Mode –
In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. The half-duplex mode is used in cases where there is no need for communication in both directions at the same time. The entire capacity of the channel can be utilized for each direction.
Example: Walkie-talkie in which message is sent one at a time and messages are sent in both directions.

Channel capacity=Bandwidth * Propagation Delay



Advantages:

- Half-duplex mode allows for bidirectional communication, which is useful in situations where devices need to send and receive data.
- It is a more efficient mode of communication than simplex mode, as the channel can be used for both transmission and reception.
- Half-duplex mode is less expensive than full-duplex mode, as it only requires one communication channel.

Disadvantages:

- Half-duplex mode is less reliable than Full-Duplex mode, as both devices cannot transmit at the same time.
- There is a delay between transmission and reception, which can cause problems in some applications.
- There is a need for coordination between the transmitting and receiving devices, which can complicate the communication process.

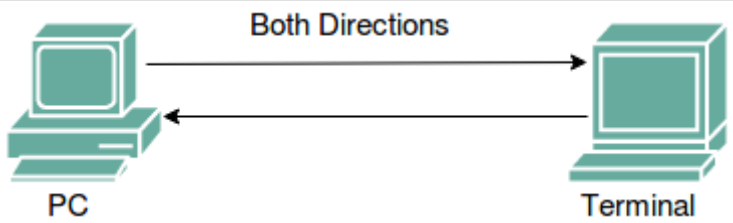
3. Full-Duplex Mode –

In full-duplex mode, both stations can transmit and receive simultaneously. In full_duplex mode, signals going in one direction share the capacity of the link with signals going in another direction, this sharing can occur in two ways:

- Either the link must contain two physically separate transmission paths, one for sending and the other for receiving.
- Or the capacity is divided between signals traveling in both directions.

Full-duplex mode is used when communication in both directions is required all the time. The capacity of the channel, however, must be divided between the two directions.
Example: Telephone Network in which there is communication between two persons by a telephone line, through which both can talk and listen at the same time.

Channel Capacity=2* Bandwidth*propagation Delay



Advantages:

- Full-duplex mode allows for simultaneous bidirectional communication, which is ideal for real-time applications such as video conferencing or online gaming.
- It is the most efficient mode of communication, as both devices can transmit and receive data simultaneously.
- Full-duplex mode provides a high level of reliability and accuracy, as there is no need for error correction mechanisms.

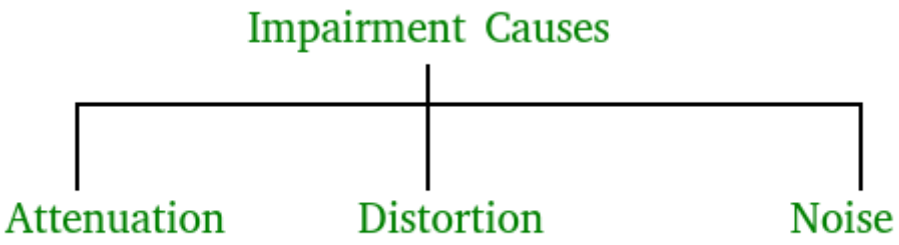
Disadvantages:

- Full-duplex mode is the most expensive mode, as it requires two communication channels.
- It is more complex than simplex and half-duplex modes, as it requires two physically separate transmission paths or a division of channel capacity.
- Full-duplex mode may not be suitable for all applications, as it requires a high level of bandwidth and may not be necessary for some types of communication.

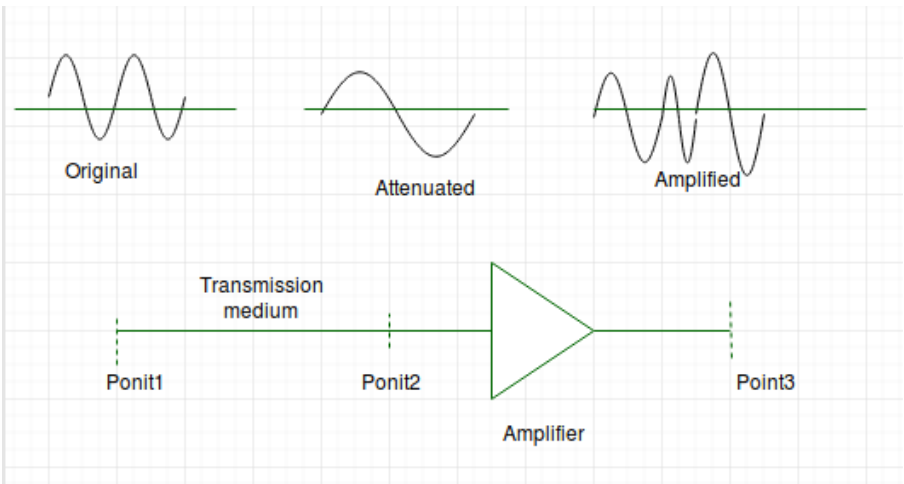
❖ Transmission Impairment in Data Communication

In communication system, analog signals travel through transmission media, which tends to deteriorate the quality of analog signal, which means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. The imperfection causes signal impairment. Below are the causes of the impairment.

Causes of impairment –



- **Attenuation –** It means loss of energy. The strength of signal decreases with increasing distance which causes loss of energy in overcoming resistance of medium. This is also known as attenuated signal. Amplifiers are used to amplify the attenuated signal which gives the original signal back and compensate for this loss.



- Image Source – aviationchief
  Attenuation is measured in **decibels(dB)**. It measures the relative strengths of two signals or one signal at two different point.

Attenuation(dB) = 10log10(P2/P1)

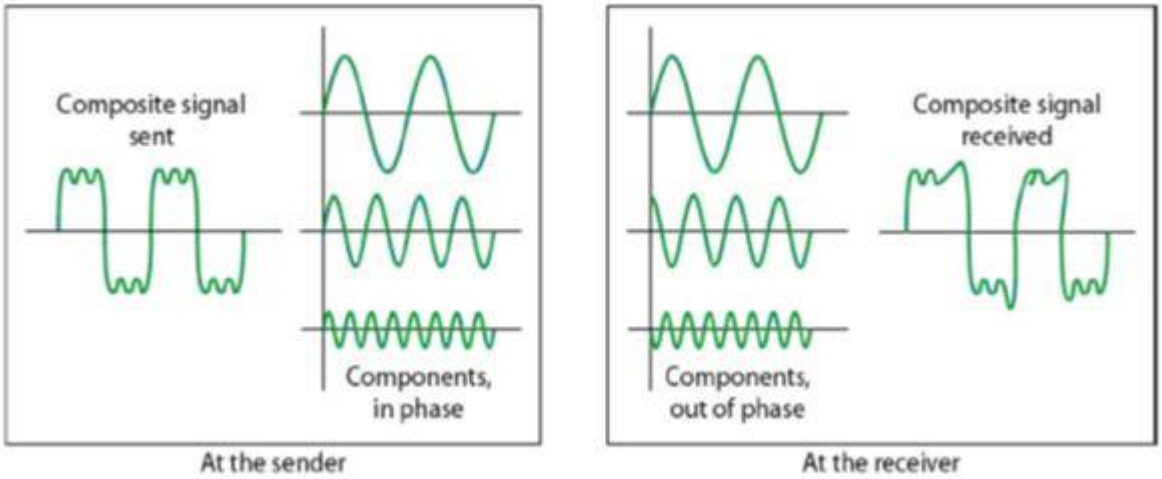
P1 is the power at sending end and P2 is the power at receiving end.

Some where the decibel is also define in terms of voltage instead of power.In this case because power is proportional to the square of the voltage the formula is

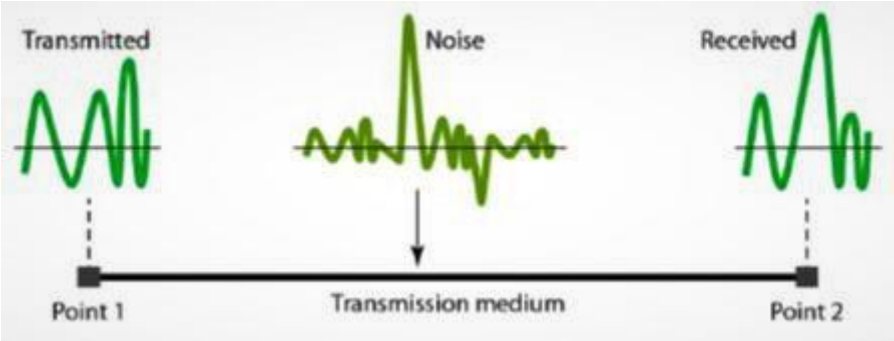
Attenuation(dB) = 20log10(V2/V1)

V1 is the voltage at sending end and V2 is the voltage at receiving end.

- **Distortion –** It means changes in the form or shape of the signal. This is generally seen in composite signals made up with different frequencies. Each frequency component has its own propagation speed travelling through a medium. And thats why it delay in arriving at the final destination Every component arrive at different time which leads to distortion. Therefore, they have different phases at receiver end from what they had at senders end.



- **Noise –** The random or unwanted signal that mixes up with the original signal is called noise. There are several types of noise such as induced noise, crosstalk noise, thermal noise and impulse noise which may corrupt the signal.
  Induced noise comes from sources such as motors and appliances. These devices act as sending antenna and transmission medium act as receiving antenna. **Thermal** noise is movement of electrons in wire which creates an extra signal. **Crosstalk** noise is when one wire affects the other wire. **Impulse** noise is a signal with high energy that comes from lightning or power lines



- To find the theoretical bit rate limit, we need to know the ration .The signal-to-noise ratio is defined as

SNR = AVG SIGNAL POWER / AVG NOISE POWER
SNR_{dB} = 10Log10SNR

EXAMPLE
The values of SNR and SNR_{dB} for a noiseless channel are
SNR = Signal Power/0 = ∞
SNR_{dB} = $10Log_{10}$ ∞ = ∞

❖ Data Rate Limits and Nyquist Bit Rate

Data rate governs the speed of data transmission. A very important consideration in data communication is how fast we can send data, in bits per second, over a channel. Data rate depends upon 3 factors:

- The bandwidth available
- Number of levels in digital signal
- The quality of the channel – level of noise

Two theoretical formulas were developed to calculate the data rate: one by Nyquist for a noiseless channel, another by Shannon for a noisy channel.

1. Noiseless Channel: Nyquist Bit Rate: For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate
Nyquist proved that if an arbitrary signal has been run through a low-pass filter of bandwidth, the filtered signal can be completely reconstructed by making only 2*Bandwidth (exact) samples per second. Sampling the line faster than 2*Bandwidth times per second is pointless because the higher-frequency components that such sampling could recover have already been filtered out. If the signal consists of L discrete levels, Nyquist's theorem states:

BitRate = 2 * Bandwidth * $\log_2(L)$ bits/sec

In the above equation, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the bit rate in bits per second.
Bandwidth is a fixed quantity, so it cannot be changed. Hence, the data rate is directly proportional to the number of signal levels.

Note: *Increasing the levels of a signal may reduce the reliability of the system.*

Examples:

Input1: Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. What can be the maximum bit rate?
Output1 : BitRate = 2 * 3000 * $\log_2(2)$ = 6000bps
Input2: We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?
Output2 : 265000 = 2 * 20000 * $\log_2(L)$
$\log_2(L) = 6.625$
$L = 2^{6.625} = 98.7$ levels

The amount of thermal noise present is measured by the ratio of the signal power to the noise power, called the ***SNR (Signal-to-Noise Ratio).***

2. Noisy Channel Shannon Capacity: In reality, we cannot have a noiseless channel; the channel is always noisy. Shannon capacity is used, to determine the theoretical highest data rate for a noisy channel:

Capacity = bandwidth * $\log_2(1 + SNR)$ bits/sec

In the above equation, bandwidth is the bandwidth of the channel, SNR is the signal-to-noise ratio, and capacity is the capacity of the channel in bits per second. Bandwidth is a fixed quantity, so it cannot be changed. Hence, the channel capacity is directly proportional to the power of the signal, as SNR = (Power of signal) / (power of noise).

The signal-to-noise ratio (S/N) is usually expressed in decibels (dB) given by the formula:

$10 * \log_{10}(S/N)$

So for example a signal-to-noise ratio of 1000 is commonly expressed as:

$10 * \log_{10}(1000) = 30$ dB.

Examples:

Input1 : A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communication. The SNR is usually 3162. What will be the capacity for this channel?

Output1 : $C = 3000 * \log_2(1 + SNR) = 3000 * 11.62 = 34860$ bps

Input2 : The SNR is often given in decibels. Assume that SNR(dB) is 36 and the channel bandwidth is 2 MHz. Calculate the theoretical channel capacity.

Output2 : $SNR(dB) = 10 * \log_{10}(SNR)$

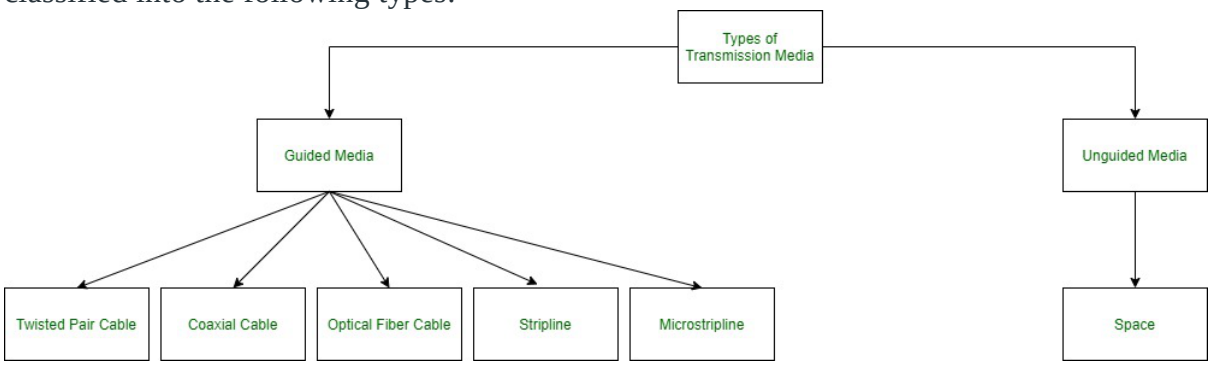
$SNR = 10^{(SNR(dB)/10)}$

$SNR = 10^{3.6} = 3981$

Hence, $C = 2 * 10^6 * \log_2(3982) = 24$ MHz

❖ Transmission Media

A transmission medium is a physical path between the transmitter and the receiver i.e. it is the channel through which data is sent from one place to another. Transmission Media is broadly classified into the following types:



1. Guided Media

Guided Media is also referred to as Wired or Bounded transmission media. Signals being transmitted are directed and confined in a narrow pathway by using physical links.

Features:

- High Speed
- Secure
- Used for comparatively shorter distances

There are 3 major types of Guided Media:

Twisted Pair Cable

It consists of 2 separately insulated conductor wires wound about each other. Generally, several such pairs are bundled together in a protective sheath. They are the most widely used Transmission Media. Twisted Pair is of two types:

- **Unshielded Twisted Pair (UTP):** UTP consists of two insulated copper wires twisted around one another. This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.

Unshielded Twisted Pair

Advantages of Unshielded Twisted Pair

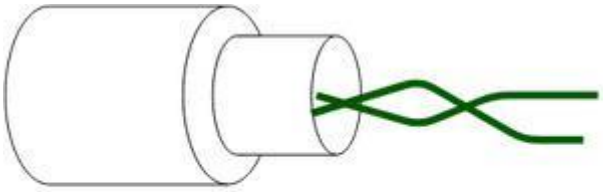
Least expensive

- Easy to install
- High-speed capacity

Disadvantages of Unshielded Twisted Pair

Susceptible to external interference

- Lower capacity and performance in comparison to STP
- Short distance transmission due to attenuation



Shielded Twisted Pair

Shielded Twisted Pair (STP): This type of cable consists of a special jacket (a copper braid covering or a foil shield) to block external interference. It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.

Advantages of Shielded Twisted Pair

- Better performance at a higher data rate in comparison to UTP
- Eliminates crosstalk
- Comparatively faster

Disadvantages of Shielded Twisted Pair

- Comparatively difficult to install and manufacture
- More expensive
- Bulky

Coaxial Cable

It has an outer plastic covering containing an insulation layer made of PVC or Teflon and 2 parallel conductors each having a separate insulated protection cover. The coaxial cable transmits information in two modes: Baseband mode(dedicated cable bandwidth) and Broadband mode(cable bandwidth is split into separate ranges). Cable TVs and analog television networks widely use Coaxial cables.

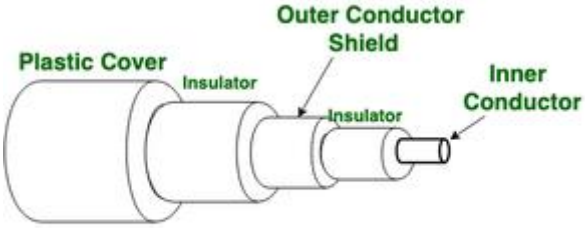


Figure of Coaxial Cable

Advantages of Coaxial Cable
Coaxial cables support high bandwidth.
- It is easy to install coaxial cables.
- Coaxial cables have better cut-through resistance so they are more reliable and durable.
- Less affected by noise or cross-talk or electromagnetic inference.
- Coaxial cables support multiple channels

Disadvantages of Coaxial Cable
- Coaxial cables are expensive.
- The coaxial cable must be grounded in order to prevent any crosstalk.
- As a Coaxial cable has multiple layers it is very bulky.
- There is a chance of breaking the coaxial cable and attaching a "t-joint" by hackers, this compromises the security of the data.

Optical Fiber Cable
Optical Fibre Cable uses the concept of refraction of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the cladding. It is used for the transmission of large volumes of data. The cable can be unidirectional or bidirectional. The WDM (Wavelength Division Multiplexer) supports two modes, namely unidirectional and bidirectional mode.

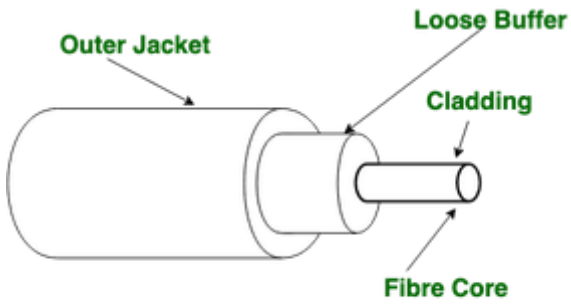


Figure of Optical Fibre Cable

Advantages of Optical Fibre Cable
- Increased capacity and bandwidth
- Lightweight
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials

Disadvantages of Optical Fibre Cable
- Difficult to install and maintain
- High cost
- Fragile

Applications of Optical Fibre Cable
- **Medical Purpose:** Used in several types of medical instruments.
- **Defence Purpose:** Used in transmission of data in aerospace.
- **For Communication:** This is largely used in formation of internet cables.
- **Industrial Purpose:** Used for lighting purposes and safety measures in designing the interior and exterior of automobiles.

Stripline
Stripline is a transverse electromagnetic (TEM) transmission line medium invented by Robert M. Barrett of the Air Force Cambridge Research Centre in the 1950s. Stripline is the earliest form of the planar transmission line. It uses a conducting material to transmit high-frequency waves it is also called a waveguide. This conducting material is sandwiched between two layers of the ground plane which are usually shorted to provide EMI immunity.

Microstripline
In this, the conducting material is separated from the ground plane by a layer of dielectric.

2. Unguided Media

It is also referred to as Wireless or Unbounded transmission media. No physical medium is required for the transmission of electromagnetic signals.

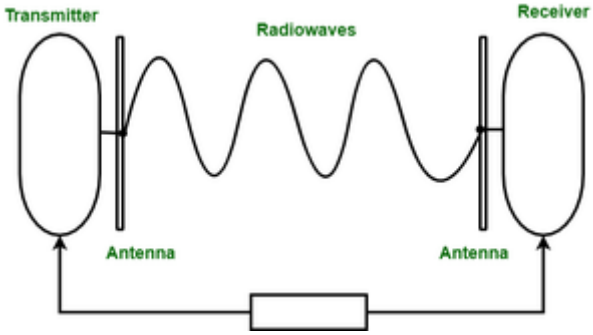
Features of Unguided Media

- The signal is broadcasted through air
- Less Secure
- Used for larger distances

There are 3 types of Signals transmitted through unguided media:

Radio Waves

Radio waves are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned. Frequency Range:3KHz – 1GHz. AM and FM radios and cordless phones use Radio waves for transmission.



Further Categorized as Terrestrial and Satellite.

Microwaves

It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range:1GHz – 300GHz. **Micro waves** are majorly used for mobile phone communication and television distribution.

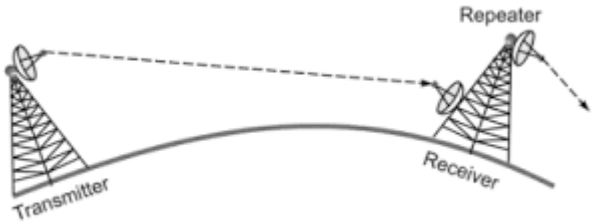
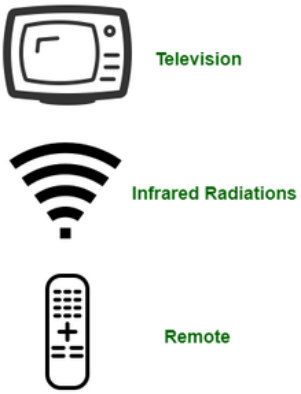


Fig: Microwave Transmission

Infrared

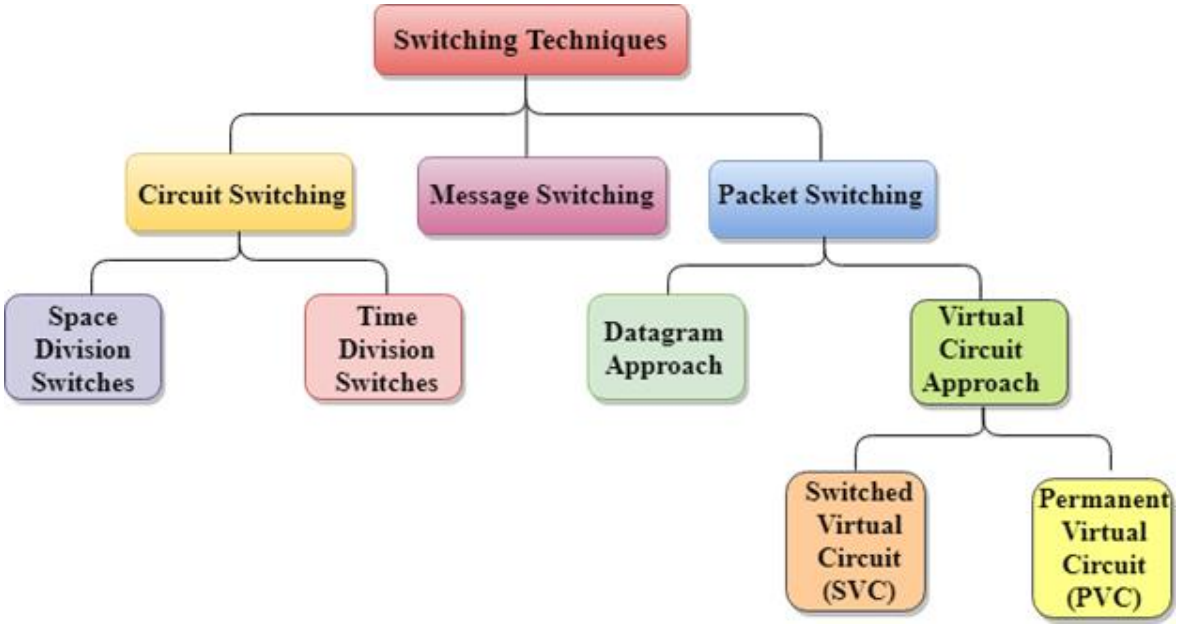
Infrared waves are used for very short distance communication. They cannot penetrate through obstacles. This prevents interference between systems. Frequency Range:300GHz – 400THz. It is used in TV remotes, wireless mouse, keyboard, printer, etc.



❖ Switching Techniques:

In large networks, there can be multiple paths from sender to receiver. The switching technique will decide the best route for data transmission. Switching technique is used to connect the systems for making one-to-one communication.

Classification of Switching Techniques

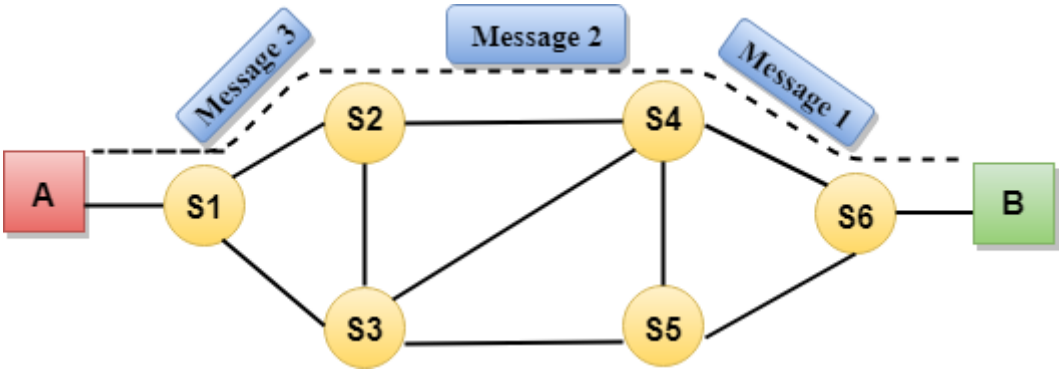


Circuit Switching

- Circuit switching is a switching technique that establishes a dedicated path between sender and receiver.
- In the Circuit Switching Technique, once the connection is established then the dedicated path will remain to exist until the connection is terminated.
- Circuit switching in a network operates in a similar way as the telephone works.
- A complete end-to-end path must exist before the communication takes place.
- In case of circuit switching technique, when any user wants to send the data, voice, video, a request signal is sent to the receiver then the receiver sends back the acknowledgment to ensure the availability of the dedicated path. After receiving the acknowledgment, dedicated path transfers the data.
- Circuit switching is used in public telephone network. It is used for voice transmission.
- Fixed data can be transferred at a time in circuit switching technology.

Communication through circuit switching has 3 phases:

- Circuit establishment
- Data transfer
- Circuit Disconnect



Circuit Switching can use either of the two technologies:

Space Division Switches:

- Space Division Switching is a circuit switching technology in which a single transmission path is accomplished in a switch by using a physically separate set of crosspoints.
- Space Division Switching can be achieved by using crossbar switch. A crossbar switch is a metallic crosspoint or semiconductor gate that can be enabled or disabled by a control unit.
- The Crossbar switch is made by using the semiconductor. For example, Xilinx crossbar switch using FPGAs.
- Space Division Switching has high speed, high capacity, and nonblocking switches.

Space Division Switches can be categorized in two ways:

- **Crossbar Switch**
- **Multistage Switch**

Crossbar Switch

The Crossbar switch is a switch that has n input lines and n output lines. The crossbar switch has n^2 intersection points known as **cross points.**

Disadvantage of Crossbar switch:

The number of cross points increases as the number of stations is increased. Therefore, it becomes very expensive for a large switch. The solution to this is to use a multistage switch.

Multistage Switch

- Multistage Switch is made by splitting the crossbar switch into the smaller units and then interconnecting them.
- It reduces the number of cross points.
- If one path fails, then there will be an availability of another path.

Advantages of Circuit Switching:

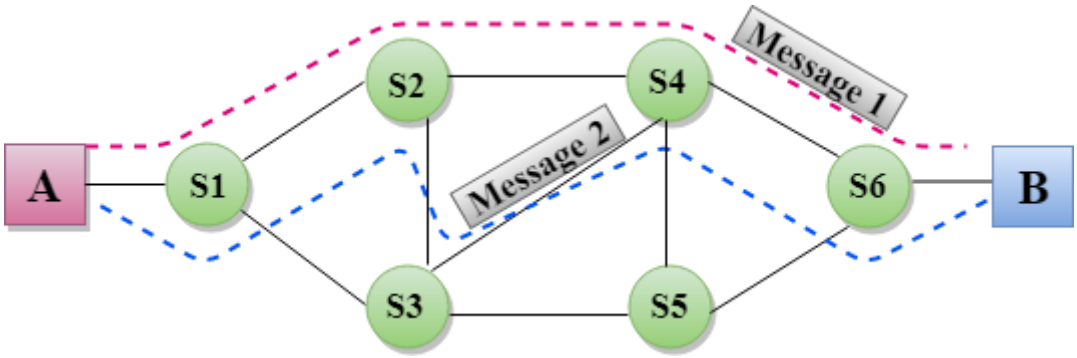
- In the case of Circuit Switching technique, the communication channel is dedicated.
- It has fixed bandwidth.

Disadvantages of Circuit Switching:

- Once the dedicated path is established, the only delay occurs in the speed of data transmission.
- It takes a long time to establish a connection approx 10 seconds during which no data can be transmitted.
- It is more expensive than other switching techniques as a dedicated path is required for each connection.
- It is inefficient to use because once the path is established and no data is transferred, then the capacity of the path is wasted.
- In this case, the connection is dedicated therefore no other data can be transferred even if the channel is free.

Message Switching

- Message Switching is a switching technique in which a message is transferred as a complete unit and routed through intermediate nodes at which it is stored and forwarded.
- In Message Switching technique, there is no establishment of a dedicated path between the sender and receiver.
- The destination address is appended to the message. Message Switching provides a dynamic routing as the message is routed through the intermediate nodes based on the information available in the message.
- Message switches are programmed in such a way so that they can provide the most efficient routes.
- Each and every node stores the entire message and then forward it to the next node. This type of network is known as **store and forward network.**
- Message switching treats each message as an independent entity.



Advantages of Message Switching

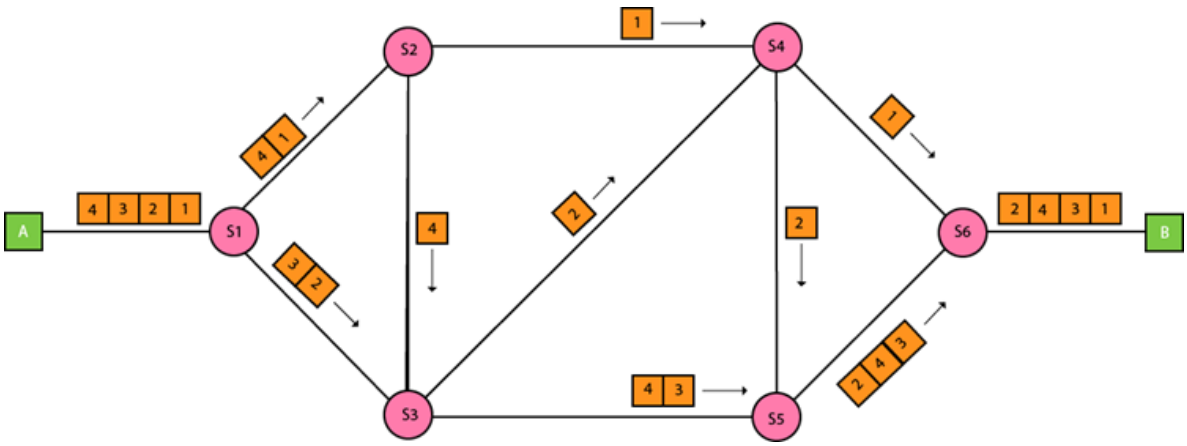
- Data channels are shared among the communicating devices that improve the efficiency of using available bandwidth.
- Traffic congestion can be reduced because the message is temporarily stored in the nodes.
- Message priority can be used to manage the network.
- The size of the message which is sent over the network can be varied. Therefore, it supports the data of unlimited size.

Disadvantages of Message Switching

- The message switches must be equipped with sufficient storage to enable them to store the messages until the message is forwarded.
- The Long delay can occur due to the storing and forwarding facility provided by the message switching technique.

Packet Switching

- The packet switching is a switching technique in which the message is sent in one go, but it is divided into smaller pieces, and they are sent individually.
- The message splits into smaller pieces known as packets and packets are given a unique number to identify their order at the receiving end.
- Every packet contains some information in its headers such as source address, destination address and sequence number.
- Packets will travel across the network, taking the shortest path as possible.
- All the packets are reassembled at the receiving end in correct order.
- If any packet is missing or corrupted, then the message will be sent to resend the message.
- If the correct order of the packets is reached, then the acknowledgment message will be sent.



Approaches of Packet Switching:

There are two approaches to Packet Switching:

Datagram Packet switching:

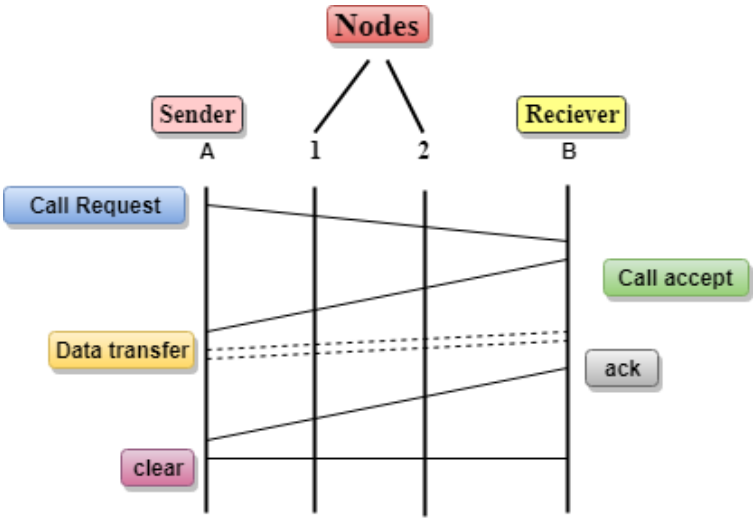
- It is a packet switching technology in which packet is known as a datagram, is considered as an independent entity. Each packet contains the information about the destination and switch uses this information to forward the packet to the correct destination.
- The packets are reassembled at the receiving end in correct order.
- In Datagram Packet Switching technique, the path is not fixed.
- Intermediate nodes take the routing decisions to forward the packets.
- Datagram Packet Switching is also known as connectionless switching.

Virtual Circuit Switching

- Virtual Circuit Switching is also known as connection-oriented switching.

- In the case of Virtual circuit switching, a preplanned route is established before the messages are sent.
- Call request and call accept packets are used to establish the connection between sender and receiver.
- In this case, the path is fixed for the duration of a logical connection.

Let's understand the concept of virtual circuit switching through a diagram:



- In the above diagram, A and B are the sender and receiver respectively. 1 and 2 are the nodes.
- Call request and call accept packets are used to establish a connection between the sender and receiver.
- When a route is established, data will be transferred.
- After transmission of data, an acknowledgment signal is sent by the receiver that the message has been received.
- If the user wants to terminate the connection, a clear signal is sent for the termination.

Advantages of Packet Switching:

- **Cost-effective:** In packet switching technique, switching devices do not require massive secondary storage to store the packets, so cost is minimized to some extent. Therefore, we can say that the packet switching technique is a cost-effective technique.
- **Reliable:** If any node is busy, then the packets can be rerouted. This ensures that the Packet Switching technique provides reliable communication.
- **Efficient:** Packet Switching is an efficient technique. It does not require any established path prior to the transmission, and many users can use the same communication channel simultaneously, hence makes use of available bandwidth very efficiently.

Disadvantages of Packet Switching:

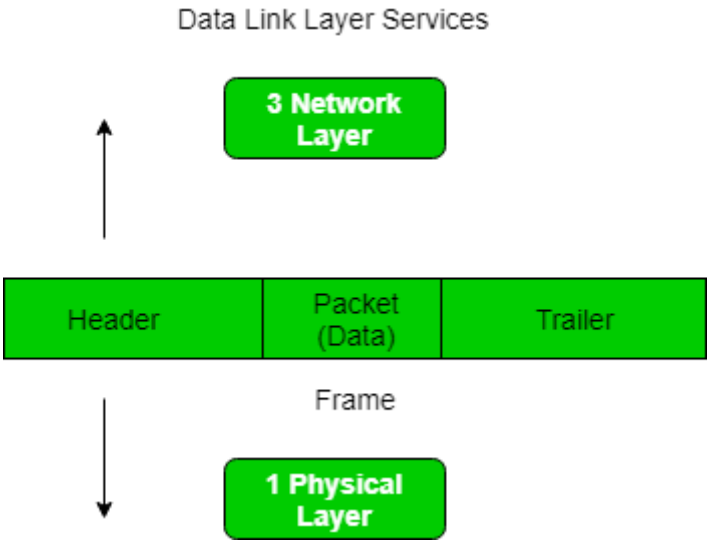
- Packet Switching technique cannot be implemented in those applications that require low delay and high-quality services.
- The protocols used in a packet switching technique are very complex and requires high implementation cost.
- If the network is overloaded or corrupted, then it requires retransmission of lost packets. It can also lead to the loss of critical information if errors are nor recovered.

❖ Data Link Layer

Framing in Data Link Layer

Frames are the units of digital transmission, particularly in computer networks and telecommunications. Frames are comparable to the packets of energy called photons in the case of light energy. Frame is continuously used in Time Division Multiplexing process.

Framing is a point-to-point connection between two computers or devices consisting of a wire in which data is transmitted as a stream of bits. However, these bits must be framed into discernible blocks of information. Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. Ethernet, token ring, frame relay, and other data link layer technologies have their own frame structures. Frames have headers that contain information such as error-checking codes.



At the data link layer, it extracts the message from the sender and provides it to the receiver by providing the sender's and receiver's addresses. The advantage of using frames is that data is broken up into recoverable chunks that can easily be checked for corruption.

The process of dividing the data into frames and reassembling it is transparent to the user and is handled by the data link layer.

Framing is an important aspect of data link layer protocol design because it allows the transmission of data to be organized and controlled, ensuring that the data is delivered accurately and efficiently.

Problems in Framing

- **Detecting start of the frame:** When a frame is transmitted, every station must be able to detect it. Station detects frames by looking out for a special sequence of bits that marks the beginning of the frame i.e. SFD (Starting Frame Delimiter).

- **How does the station detect a frame:** Every station listens to link for SFD pattern through a sequential circuit. If SFD is detected, sequential circuit alerts station. Station checks destination address to accept or reject frame.
- **Detecting end of frame:** When to stop reading the frame.
- **Handling errors:** Framing errors may occur due to noise or other transmission errors, which can cause a station to misinterpret the frame. Therefore, error detection and correction mechanisms, such as cyclic redundancy check (CRC), are used to ensure the integrity of the frame.
- **Framing overhead:** Every frame has a header and a trailer that contains control information such as source and destination address, error detection code, and other protocol-related information. This overhead reduces the available bandwidth for data transmission, especially for small-sized frames.
- **Framing incompatibility:** Different networking devices and protocols may use different framing methods, which can lead to framing incompatibility issues. For example, if a device using one framing method sends data to a device using a different framing method, the receiving device may not be able to correctly interpret the frame.
- **Framing synchronization:** Stations must be synchronized with each other to avoid collisions and ensure reliable communication. Synchronization requires that all stations agree on the frame boundaries and timing, which can be challenging in complex networks with many devices and varying traffic loads.
- **Framing efficiency:** Framing should be designed to minimize the amount of data overhead while maximizing the available bandwidth for data transmission. Inefficient framing methods can lead to lower network performance and higher latency.

Types of framing

There are two types of framing:

1. Fixed-size: The frame is of fixed size and there is no need to provide boundaries to the frame, the length of the frame itself acts as a delimiter.

- **Drawback:** It suffers from internal fragmentation if the data size is less than the frame size
- **Solution:** Padding

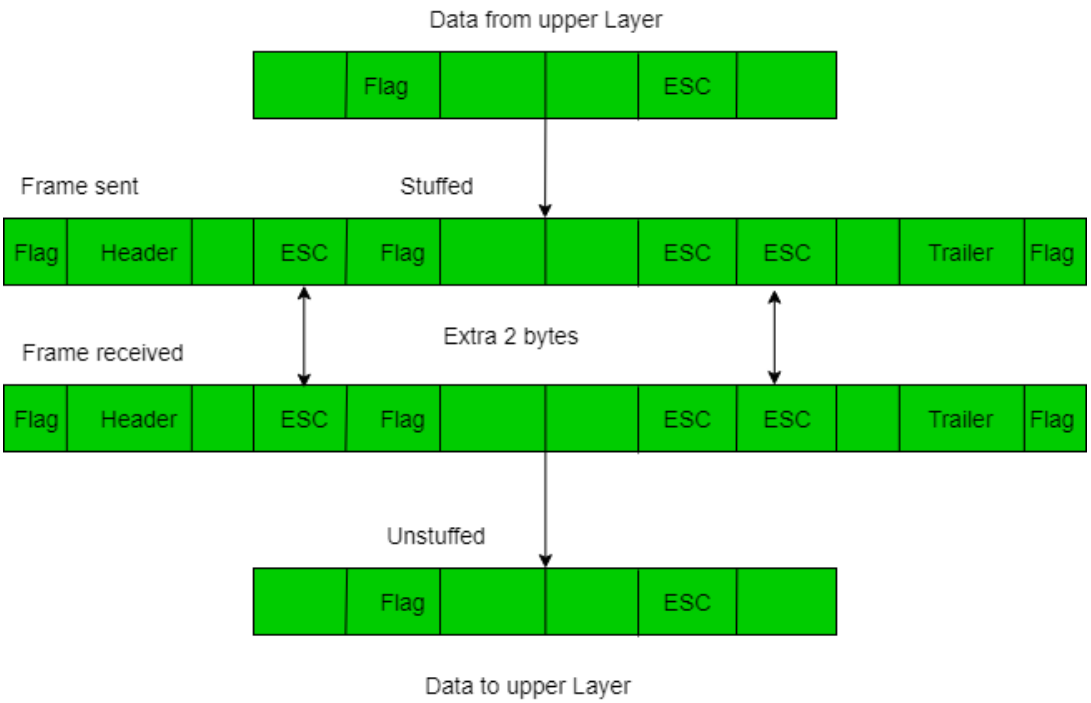
2. Variable size: In this, there is a need to define the end of the frame as well as the beginning of the next frame to distinguish. This can be done in two ways:

1. **Length field –** We can introduce a length field in the frame to indicate the length of the frame. Used in **Ethernet(802.3)**. The problem with this is that sometimes the length field might get corrupted.
2. **End Delimiter (ED) –** We can introduce an ED(pattern) to indicate the end of the frame. Used in **Token Ring**. The problem with this is that ED can occur in the data. This can be solved by:

   1. Character/Byte Stuffing: Used when frames consist of characters. If data contains ED then, a byte is stuffed into data to differentiate it from ED.

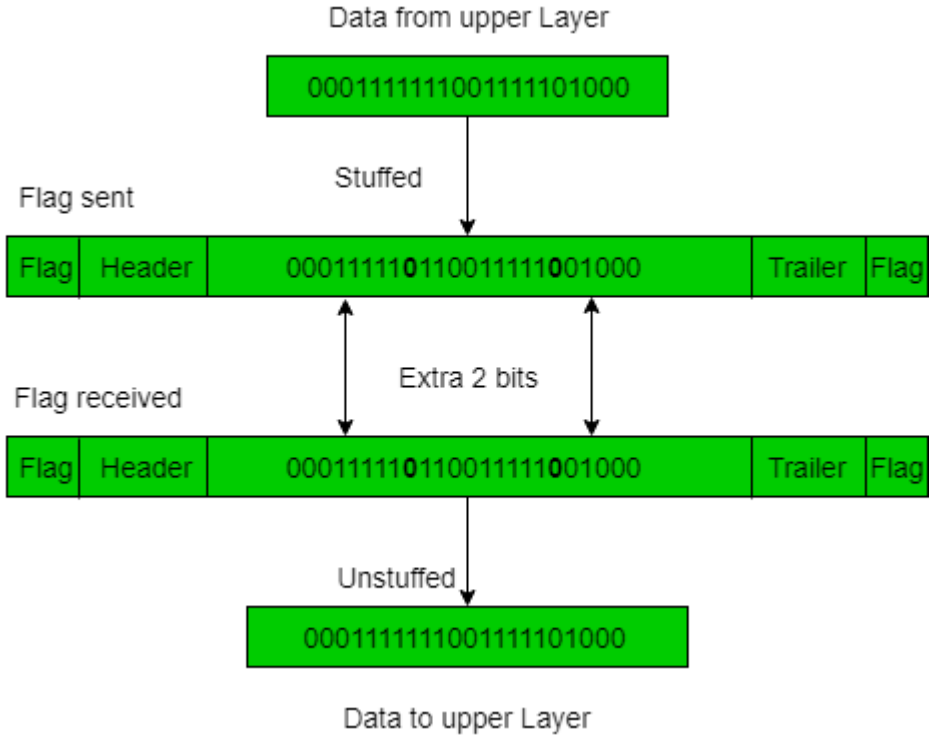
   Let ED = "$" –> if data contains '$' anywhere, it can be escaped using '\O' character.

   –> if data contains '\O$' then, use '\O\O\O$'($ is escaped using \O and \O is escaped using \O).



Disadvantage – It is very costly and obsolete method.

2. Bit Stuffing: Let ED = 01111 and if data = 01111
–> Sender stuffs a bit to break the pattern i.e. here appends a 0 in data = 0111**0**1.
–> Receiver receives the frame.
–> If data contains <u>0111</u>01, receiver removes the 0 and reads the data.



Examples:

- If Data –> 011100011110 and ED –> 0111 then, find data after bit stuffing.

--> 011**0**10001**1**01**1**0**0**

- If Data –> 110001001 and ED –> 1000 then, find data after bit stuffing?

--> 1100**1**0100**1**1

framing in the Data Link Layer also presents some challenges, which include:

Variable frame length: The length of frames can vary depending on the data being transmitted, which can lead to inefficiencies in transmission. To address this issue, protocols such as HDLC and PPP use a flag sequence to mark the start and end of each frame.
Bit stuffing: Bit stuffing is a technique used to prevent data from being interpreted as control characters by inserting extra bits into the data stream. However, bit stuffing can lead to issues with synchronization and increase the overhead of the transmission.
Synchronization: Synchronization is critical for ensuring that data frames are transmitted and received correctly. However, synchronization can be challenging, particularly in high-speed networks where frames are transmitted rapidly.
Error detection: Data Link Layer protocols use various techniques to detect errors in the transmitted data, such as checksums and CRCs. However, these techniques are not foolproof and can miss some types of errors.
Efficiency: Efficient use of available bandwidth is critical for ensuring that data is transmitted quickly and reliably. However, the overhead associated with framing and error detection can reduce the overall efficiency of the transmission.

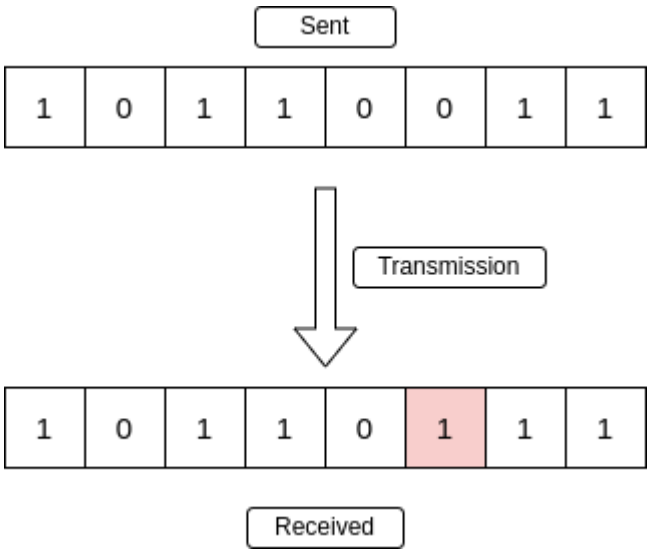
❖ Error Detection in Computer Networks

Error is a condition when the receiver's information does not match the sender's. Digital signals suffer from noise during transmission that can introduce errors in the binary bits traveling from sender to receiver. That means a 0 bit may change to 1 or a 1 bit may change to 0.
Data (Implemented either at the Data link layer or Transport Layer of the OSI Model) may get scrambled by noise or get corrupted whenever a message is transmitted. To prevent such errors, error-detection codes are added as extra data to digital messages. This helps in detecting any errors that may have occurred during message transmission.

Types of Errors

Single-Bit Error

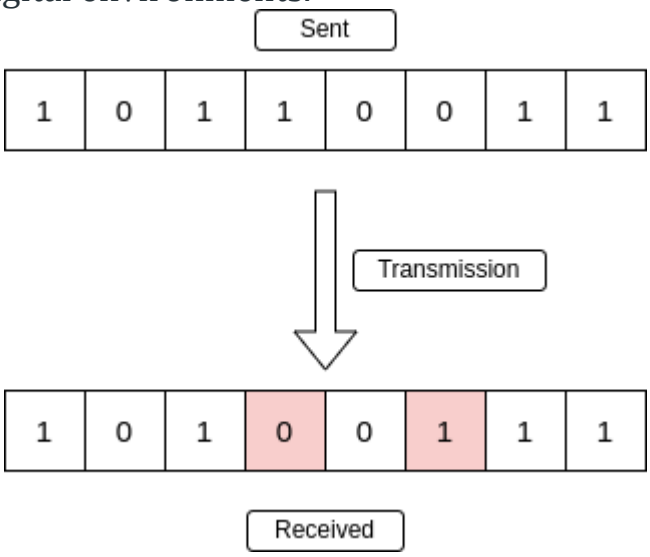
A single-bit error refers to a type of data transmission error that occurs when one bit (i.e., a single binary digit) of a transmitted data unit is altered during transmission, resulting in an incorrect or corrupted data unit.



Single-Bit Error

Multiple-Bit Error

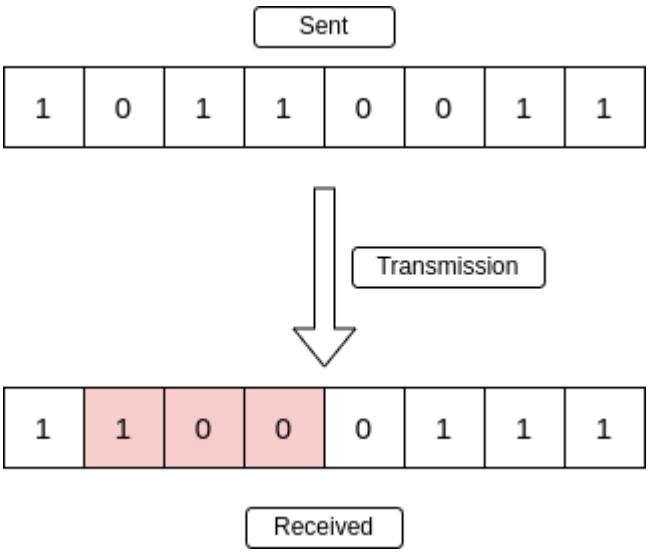
A multiple-bit error is an error type that arises when more than one bit in a data transmission is affected. Although multiple-bit errors are relatively rare when compared to single-bit errors, they can still occur, particularly in high-noise or high-interference digital environments.



Multiple-Bit Error

Burst Error

When several consecutive bits are flipped mistakenly in digital transmission, it creates a burst error. This error causes a sequence of consecutive incorrect values.



Burst Error

Error Detection Methods

To detect errors, a common technique is to introduce redundancy bits that provide additional information. Various techniques for error detection include:

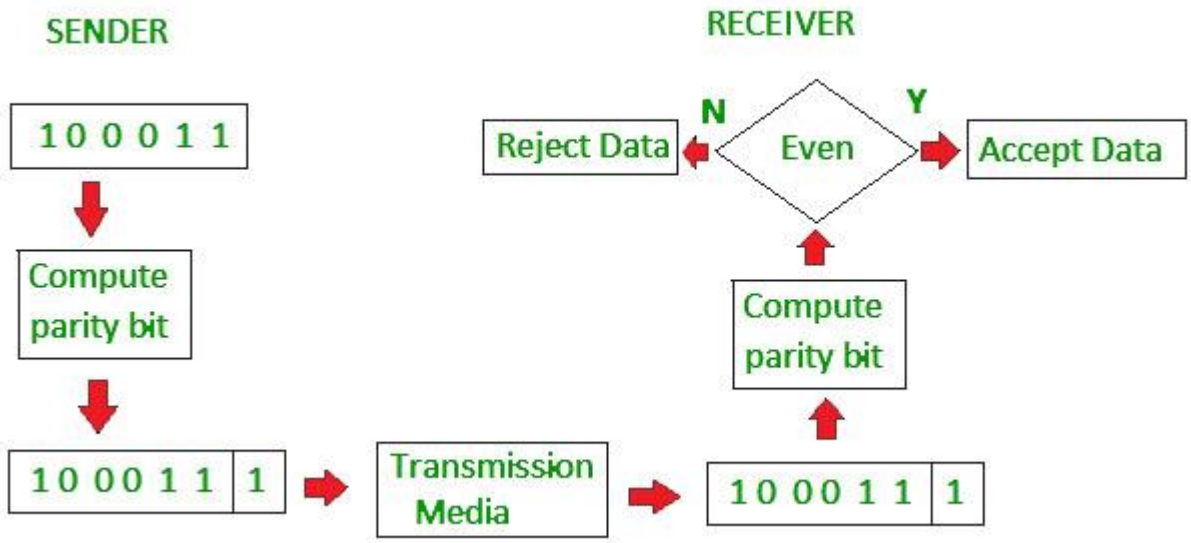
- Simple Parity Check
- Two-Dimensional Parity Check
- Checksum
- Cyclic Redundancy Check (CRC)

Simple Parity Check

Simple-bit parity is a simple error detection method that involves adding an extra bit to a data transmission. It works as:

- 1 is added to the block if it contains an odd number of 1's, and
- 0 is added if it contains an even number of 1's

This scheme makes the total number of 1's even, that is why it is called [even parity] checking.



Advantages of Simple Parity Check

- Simple parity check can detect all single bit error.
- Simple parity check can detect an odd number of errors.
- **Implementation**: Simple Parity Check is easy to implement in both hardware and software.
- **Minimal Extra Data**: Only one additional bit (the parity bit) is added per data unit (e.g., per byte).
- **Fast Error Detection**: The process of calculating and checking the parity bit is quick, which allows for rapid error detection without significant delay in data processing or communication.
- **Single-Bit Error Detection**: It can effectively detect single-bit errors within a data unit, providing a basic level of error detection for relatively low-error environments.

Disadvantages of Simple Parity Check

- Single Parity check is not able to detect even no. of bit error.
- **For example,** the Data to be transmitted is **101010**. Codeword transmitted to the receiver is 1010101 (we have used even parity). Let's assume that during transmission, two of the bits of code word flipped to 1111101.
  On receiving the code word, the receiver finds the no. of ones to be even and hence **no error,** *which is a wrong assumption.*

Two-Dimensional Parity Check

Two-dimensional Parity check bits are calculated for each row, which is equivalent to a simple parity check bit. Parity check bits are also calculated for all columns, then both are sent along with the data. At the receiving end, these are compared with the parity bits calculated on the received data.

Original Data

10011001	11100010	00100100	10000100

Row parities

10011001	0
11100010	0
00100100	0
10000100	0
11011011	0

Column parities

100110010	111000100	001001000	100001000	110110110

Data to be sent

Advantages of Two-Dimensional Parity Check

- Two-Dimensional Parity Check can detect and correct all single bit error.
- Two-Dimensional Parity Check can detect two or three bit error that occur any where in the matrix.

Disadvantages of Two-Dimensional Parity Check

- Two-Dimensional Parity Check can not correct two or three bit error. It can only detect two or three bit error.
- If we have a error in the parity bit then this scheme will not work.

Checksum

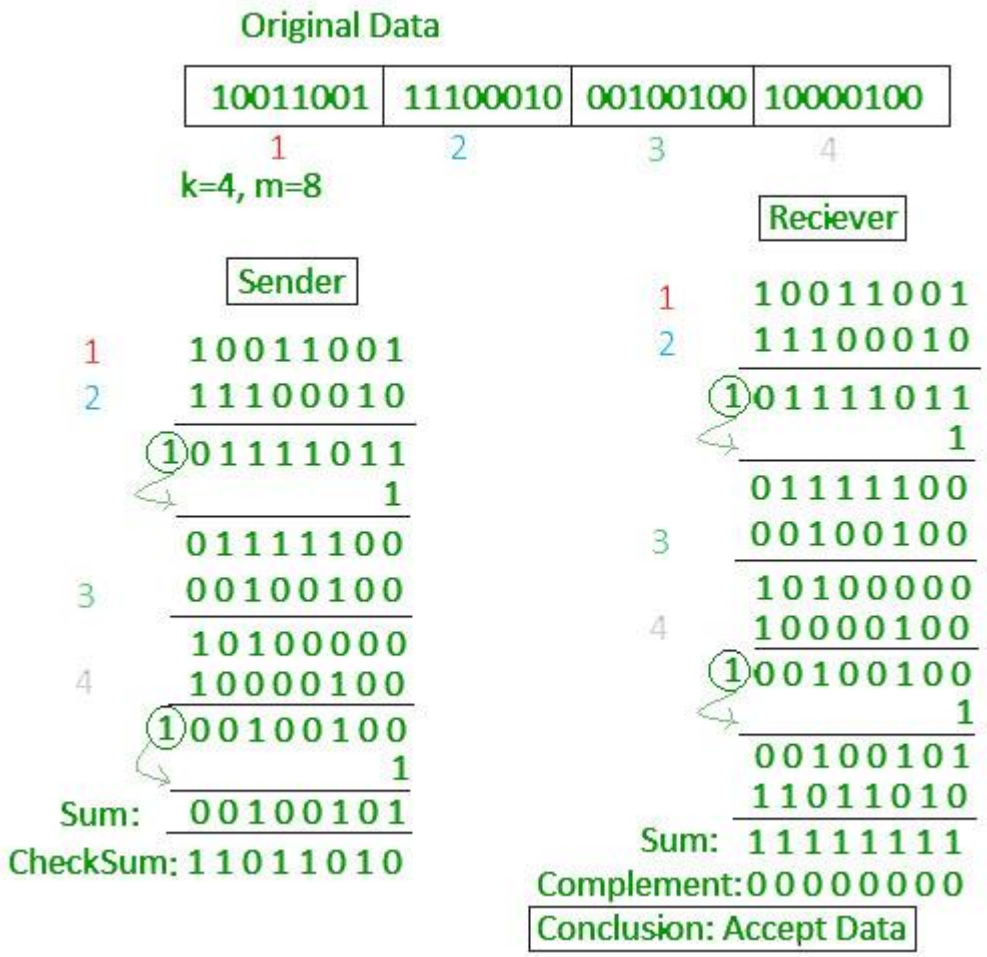
Checksum error detection is a method used to identify errors in transmitted data. The process involves dividing the data into equally sized segments and using a 1's complement to calculate the sum of these segments. The calculated sum is then sent along with the data to the receiver. At the receiver's end, the same process is repeated and if all zeroes are obtained in the sum, it means that the data is correct.

Checksum – Operation at Sender's Side

- Firstly, the data is divided into k segments each of m bits.
- On the sender's end, the segments are added using 1's complement arithmetic to get the sum. The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.

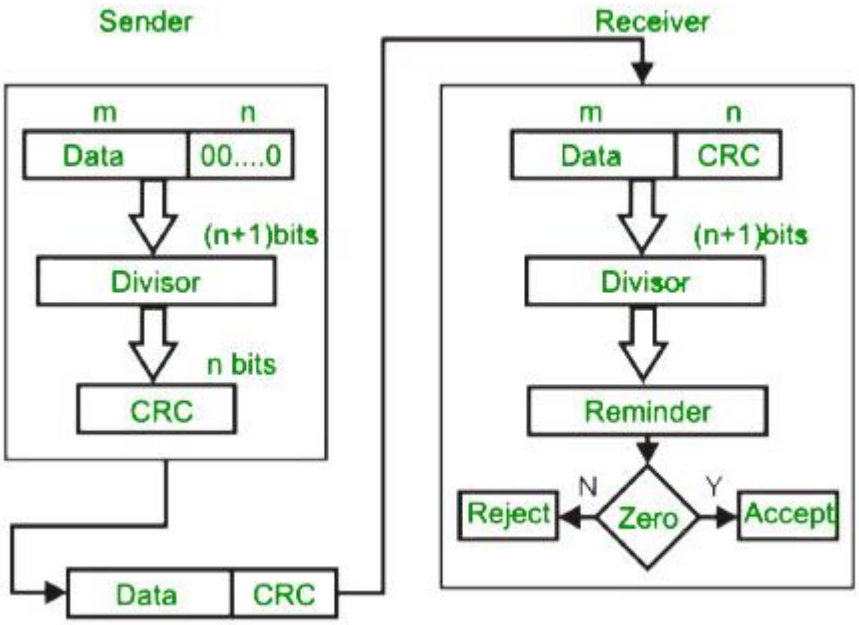
Checksum – Operation at Receiver's Side

- At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.



Cyclic Redundancy Check (CRC)

- Unlike the checksum scheme, which is based on addition, CRC is based on binary division.
- In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of the data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number.
- At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
- A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.



CRC Working

We have given dataword of length n and divisor of length k.

Step 1: Append (k-1) zero's to the original message

Step 2: Perform modulo 2 division

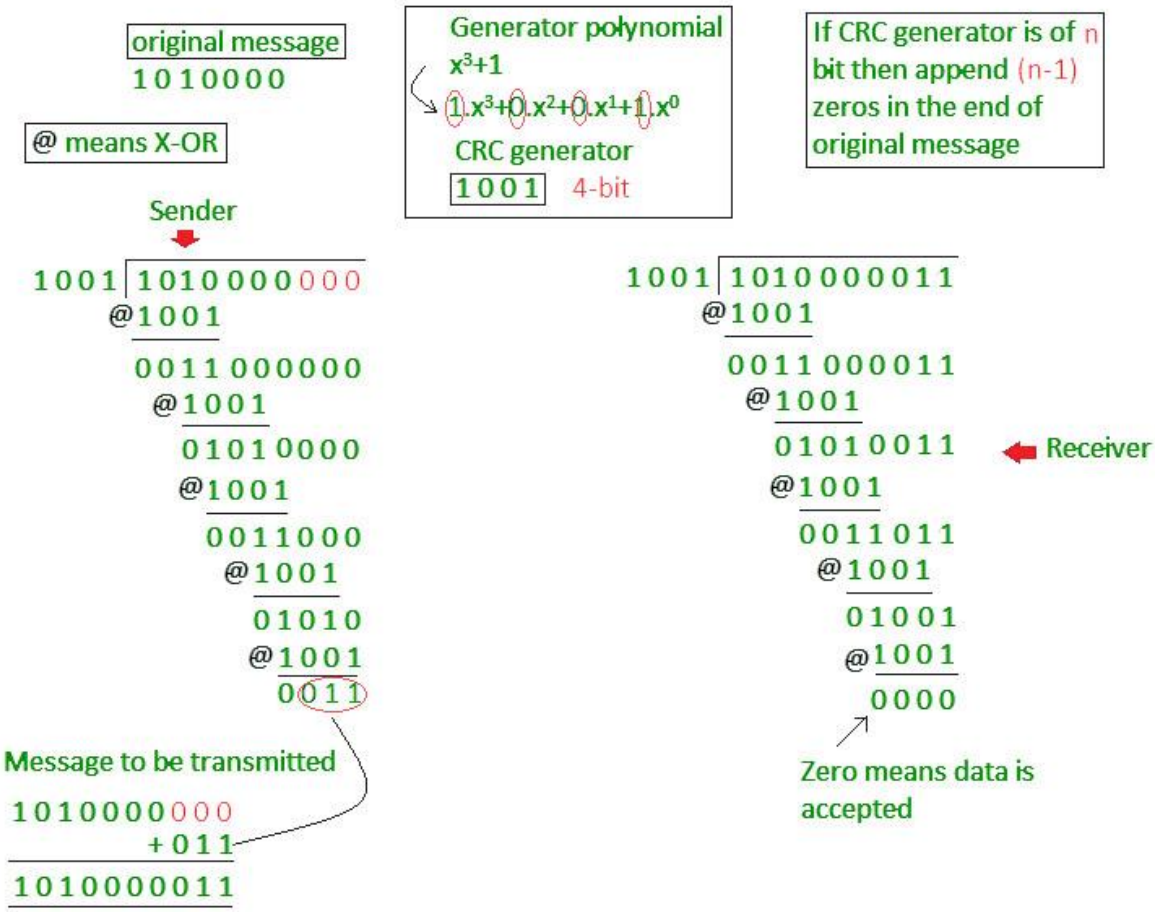
Step 3: Remainder of division = CRC

Step 4: Code word = Data with append k-1 zero's + CRC

Note:

- CRC must be k-1 bits
- Length of Code word = n+k-1 bits

Example: Let's data to be send is 1010000 and divisor in the form of polynomial is x3+1. CRC method discussed below.



Advantages of Error Detection

- **Increased Data Reliability:** Error detection ensures that the data transmitted over the network is reliable, accurate, and free from errors. This ensures that the recipient receives the same data that was transmitted by the sender.
- **Improved Network Performance:** Error detection mechanisms can help to identify and isolate network issues that are causing errors. This can help to improve the overall performance of the network and reduce downtime.
- **Enhanced Data Security:** Error detection can also help to ensure that the data transmitted over the network is secure and has not been tampered with.

Disadvantages of Error Detection

- **Overhead**: Error detection requires additional resources and processing power, which can lead to increased overhead on the network. This can result in slower network performance and increased latency.
- **False Positives:** Error detection mechanisms can sometimes generate false positives, which can result in unnecessary retransmission of data. This can further increase the overhead on the network.
- **Limited Error Correction:** Error detection can only identify errors but cannot correct them. This means that the recipient must rely on the sender to retransmit the data, which can lead to further delays and increased network overhead.

What are Errors in Networking?

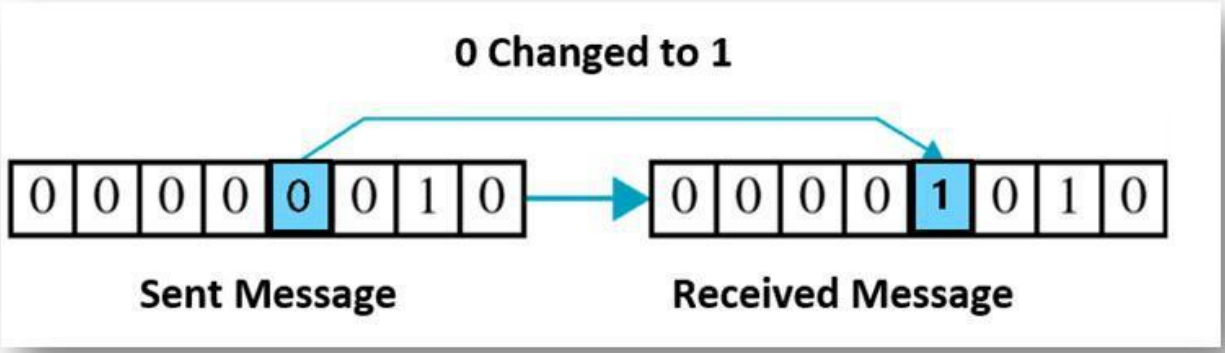
The error simply means any flaw or deviation that occurs while the information is transmitted from the source to the destination in a computer network. In other words, if the message or data transmitted by the source is not identical to the one received at the destination, we can say that there is some Error in the Computer Network.

In a Network, the data or message is transferred in the form of bits that follow a specific sequence. If there is any deviation in the bit sequence, it is called an Error. Now, let us understand the types of Errors that occur in the Compute Network.

Types of Network Errors

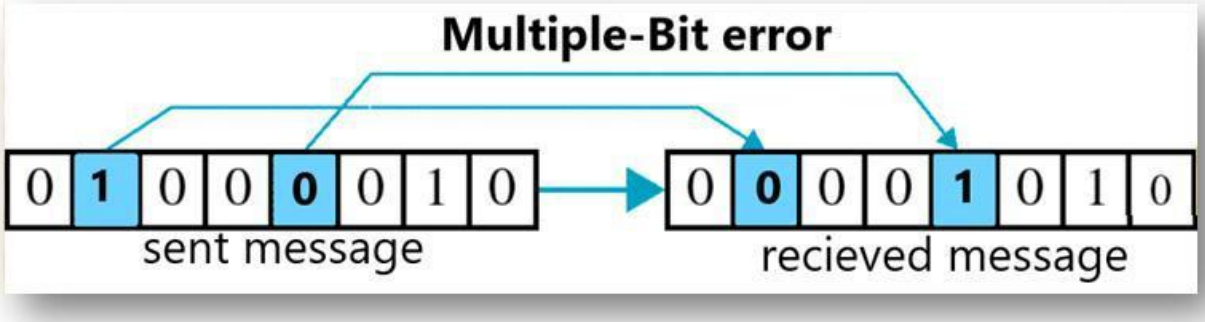
Single-Bit Error

This error occurs when the single bit (single binary digit) in the message is flipped during the transmission and the received message at the destination side differs from the original message by a single bit. For example, if a single 1(set bit) in the source message is flipped to 0 or a 0 is flipped to 1, we can say that there is a Single-Bit Error in Transmission which is shown below:



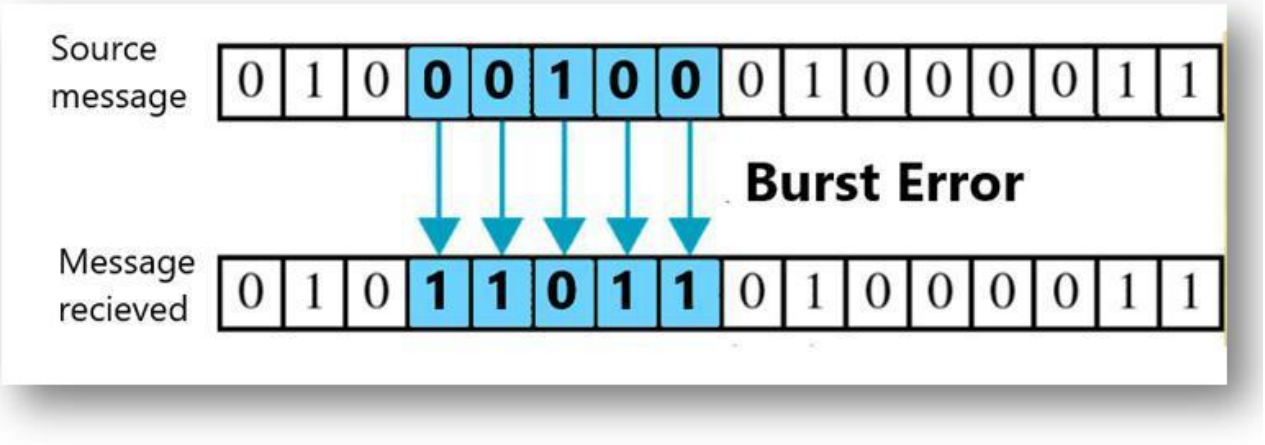
Multiple-Bit Error

This occurs when there is a deviation in multiple bits of the message during its transfer from source to destination. The bits that are flipped need not be a continuous sequence of bits. Thus, a Multiple-Bit Error is found when any non-continuous sequence of bits is flipped in the transmitted message. The example of this error is illustrated below:



Burst Error

This is similar to the Multiple-Bits Error with just one difference the bits that are flipped follow a continuous sequence in their bit representation. This error can occur due a physical damage to the disk which causes a change in the consecutive bit sequence of the source message. An example of a Burst Error is demonstrated below:

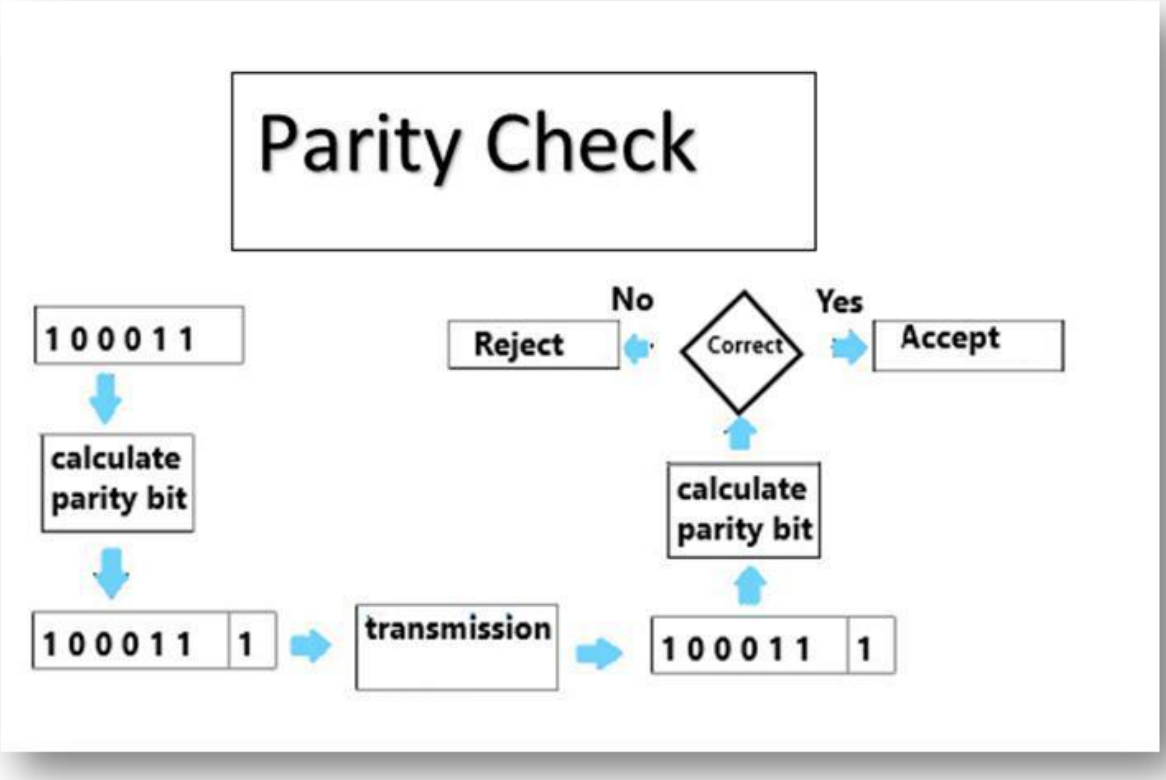


After we have learned about the different types of errors, let us see how to identify them so that we can select the appropriate method to correct the Network errors.

Error Identification in Computer Network

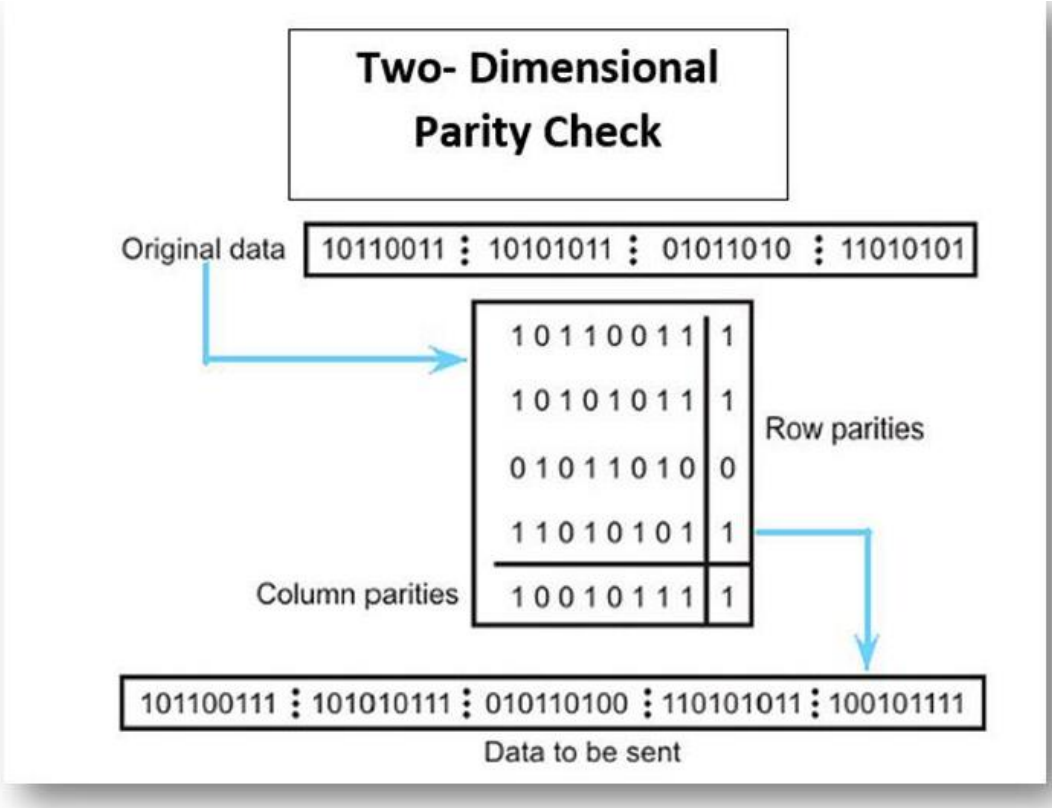
Parity Check

In this technique, the sender appends an extra parity bit to the message. If the total number of ones in the bit sequence is odd, 1 is added as a parity bit. And, if it is even, 0 acts as a parity bit. During transmission or storage, if an error alters the number of bits, the parity bit will be used to identify whether there is an error or not. This is illustrated as shown below:



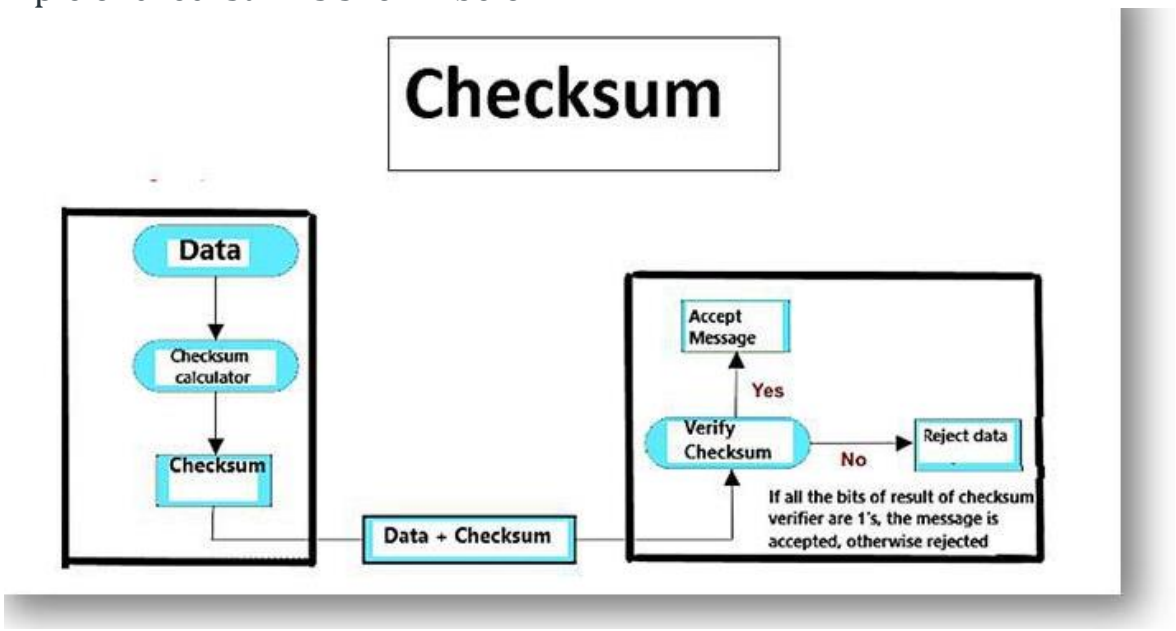
Two-dimensional Parity Check

This is just a variation of the Single-Bit Parity Check in which the original method is transformed into a matrix in which each column is associated with a parity bit based on the number of occurrences of 1s. At the receiving end, if the message has been corrupted in some way, the parity bit can identify that the message is corrupted due to some error. This is illustrated in the below example.



Checksum

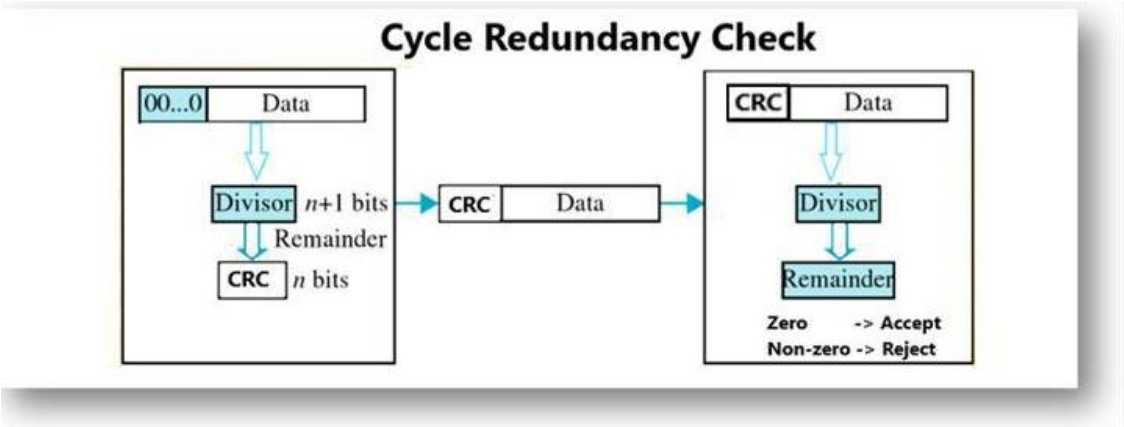
This method adds up all the bits and adds the sum to the message while transmitting. This sum is called the checksum. The sender calculates the checksum before transmitting the data, and the recipient recalculates it upon receiving the data. If the two checksum values do not match with each other, there is some error in the network. However, if they match, there is no error. An example of checksum is shown below:



Cyclic Redundancy Check (CRC)

The CRC method is similar to the Checksum method with the difference that this method deals with binary division instead of binary addition. In this method, the repeated or redundant sequence of bits in the source message is identified. Then,

it is added to the source message. Now, the source message will be easily divisible by the binary number formed from redundant bits which is called CRC Value.
Thus, the source message is divided by the CRC value which gives the divisor. Before the message is accepted, the receiver divides the bit sequence by the CRC value. If the remainder is zero, there is no error. But, if the remainder after dividing by the CRC value is not zero, it means that there is an error that needs to be corrected. The CRC method is outlined below:



After we have learned to identify the Errors in Computer Networks, it is now time to correct them.

Error Correction in Computer Networks

Once the errors are detected in the network, the deviated bits sequence needs to be replaced with the right bit sequence so that the receiver can accept the data and process it. This method is called Error Correction. We can correct the errors in the Network in two different ways which are listed below:

- Forward Error Correction: In this Error Correction Scenario, the receiving end is responsible for correcting the network error. There is no need for retransmission of the data from the sender's side.
- Backward Error Correction: Backward Error Correction means that the receiver needs to correct the error either by transmitting the corrupted message or retransmitting the entire message to the destination.

However, there is one of the most widely used Error Correction methods which is called 'Hamming Code' which was designed by R.W.Hamming. Let us have a quick look at it.

Hamming Code Error Correction

In this method, extra parity bits are appended to the message which are used by the receiver to correct the single bit error and multiple bit error. Consider the below example to understand this method in a better way.
Suppose the sender wants to transmit the message whose bit representation is '1011001.' In this message:

- Total number of bits(d) = 7
- Total of redundant bits(r) = 4 (This is because the message has four 1's in it)

- Thus, total bits(d+r) = 7 + 4 = 11

Also, by convention, the redundant bits are always placed in the places which are powers of 2. Now, this message will take the format as shown below:

R1, R2, R3, R4 are Redundant Bits

-	-	-	2^3	-	-	-	2^2	-	2^1	2^0	Power of 2
1	0	1	R4	1	0	0	R3	1	R2	R1	Message(bit sequence)
11	10	9	8	7	6	5	4	3	2	1	Total bits (d+r)

Therefore, we have R1, R2, R3, and R4 as redundant bits which will be calculated according to the following rules:

- R1 includes all the positions whose binary representation has 1 in their least significant bit. Thus, R1 covers positions 1, 3, 5, 7, 9, 11.
- R2 includes all the positions whose binary representation has 1 in the second position from the least significant bit. Thus, R2 covers positions 2,3,6,7,10,11.
- R3 includes all the positions whose binary representation has 1 in the third position from the least significant bit. Hence, R3 covers positions 4, 5, 6, 7.
- R4 includes all the positions whose binary representation has 1 in the fourth position from the least significant bit due to which R4 covers positions 8,9,10,11.

These rules are illustrated below:

Position	R4	R3	R2	R1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1

Now, we calculate the value of R1, R2, R3 and R4 as follows:

- Since the total number of 1s in all the bit positions corresponding to R1 is an even number. R1 = 0.
- Since the total number of 1s in all the bit positions corresponding to R2 is an odd number, R2= 1.
- Since the total number of 1s in all the bit positions corresponding to R3 is an odd number, R3= 1.
- Since the total number of 1s in all the bit positions corresponding to R4 is even, R4 = 0.

Therefore, the message to be transmitted becomes:

R1, R2, R3, R4 are Redundant Bits

-	-	-	2^3	-	-	-	2^2	-	2^1	2^0	Power of 2
1	0	1	0	1	0	0	1	1	1	0	Message(bit sequence)
11	10	9	8	7	6	5	4	3	2	1	Total bits (d+r)

This message is transmitted at the receiver's end. Suppose, bit 6 becomes corrupted and changes to 0. Then, the message becomes '10101101110.' So, at the receiver's end, the number of 1's in the respective bit positions of R1, R2, R3, and R4 is rechecked to correct the corrupted bit. This is done in the following steps: For all the parity bits we will check the

- For R1: bits 1, 3, 5, 7, 9, and 11 are checked. We can see that the number of 1's in these bit positions is 4(even) so R1 = 0.
- For R2: bits 2,3,6,7,10,11 are checked. You can observe that the number of 1's in these bit positions is 5(odd) so we get a R2 = 1.
- For R3: bits 4, 5, 6, and 7 are checked. We see that the number of 1's in these bit positions is 3(odd). Hence, R3 = 1.
- For R8: bits 8,9,10,11 are observed. Here, the number of 1's in these bit positions is 2 and that's even so we get R4 = 0.

If we observe the Redundant bits, they give the binary number 0110 whose decimal representation is 6. Thus, bit 6 contains an error. To correct the error the 6th bit is changed from 1 to 0 to correct the error.

Multiple Access Protocols in Computer Network

Multiple Access Protocols are methods used in computer networks to control how data is transmitted when multiple devices are trying to communicate over the same network. These protocols ensure that data packets are sent and received efficiently, without collisions or interference. They help manage the network traffic so that all devices can share the communication channel smoothly and effectively.

Who is Responsible for the Transmission of Data?

The **Data Link Layer** is responsible for the transmission of data between two nodes. Its main functions are:

- Data Link Control
- Multiple Access Control

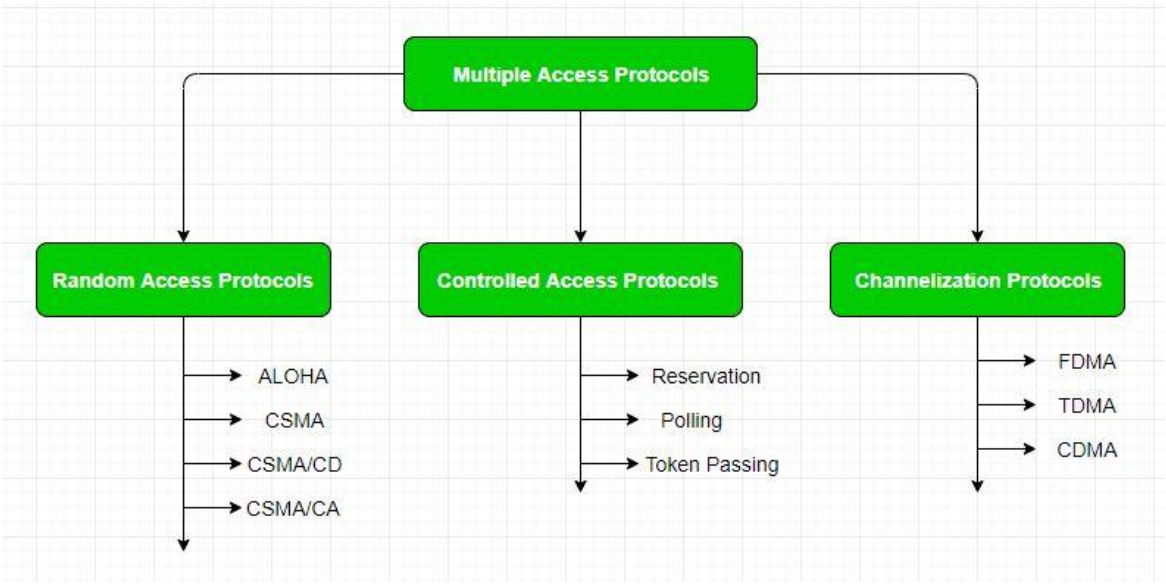
Data Link Control

The data link control is responsible for the reliable transmission of messages over transmission channels by using techniques like framing, error control and flow control.

Multiple Access Control

If there is a dedicated link between the sender and the receiver then data link control layer is sufficient, however if there is no dedicated link present then multiple stations can access the channel simultaneously. Hence multiple access protocols are required to decrease collision and avoid crosstalk. For example, in a classroom full of students, when a teacher asks a question and all the students (or stations) start answering simultaneously (send data at same time) then a lot of chaos is created (data overlap or data lost) then it is the job of the teacher (multiple access protocols) to manage the students and make them answer one at a time.

Thus, protocols are required for sharing data on non-dedicated channels. Multiple access protocols can be subdivided further as



1. Random Access Protocol

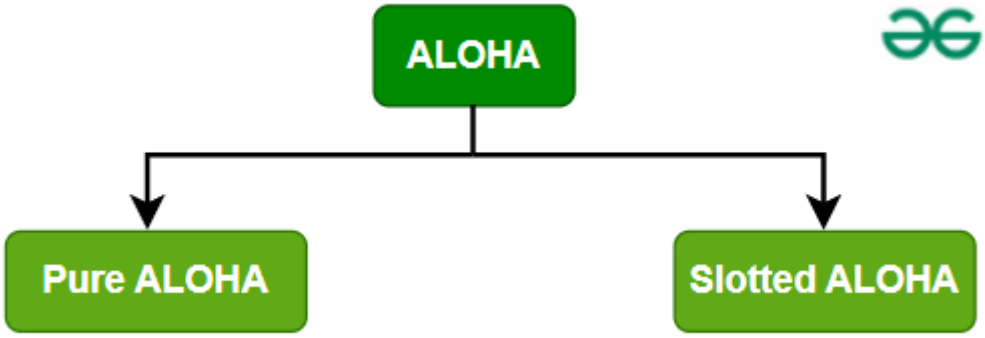
In this, all stations have same superiority that is no station has more priority than another station. Any station can send data depending on medium's state (idle or busy). It has two features:

- There is no fixed time for sending data
- There is no fixed sequence of stations sending data

The Random access protocols are further subdivided as:

ALOHA

It was designed for wireless LAN but is also applicable for shared medium. In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.

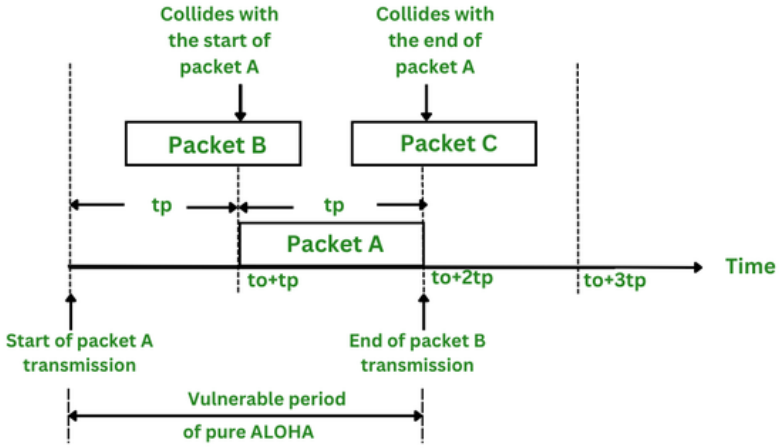


Types of Aloha

Pure ALOHA

When a station sends data it waits for an acknowledgement. If the acknowledgement doesn't come within the allotted time then the station waits for a random amount of time called back-off time (Tb) and re-sends the data. Since different stations wait for different amount of time, the probability of further collision decreases.

Vulnerable Time = 2* Frame transmission time
Throughput = G exp{-2*G}
Maximum throughput = 0.184 for G=0.5

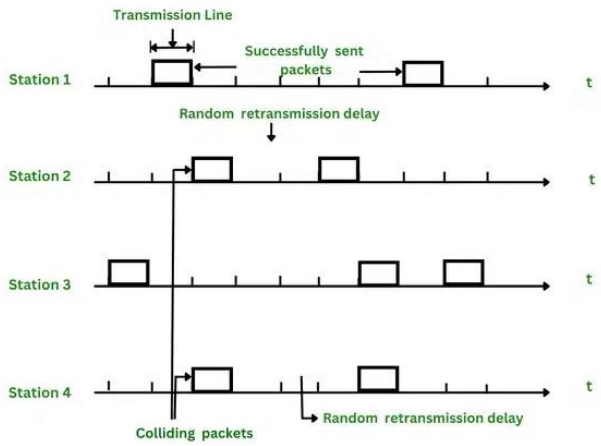


Pure Aloha

Slotted ALOHA

It is similar to pure aloha, except that we divide time into slots and sending of data is allowed only at the beginning of these slots. If a station misses out the allowed time, it must wait for the next slot. This reduces the probability of collision.

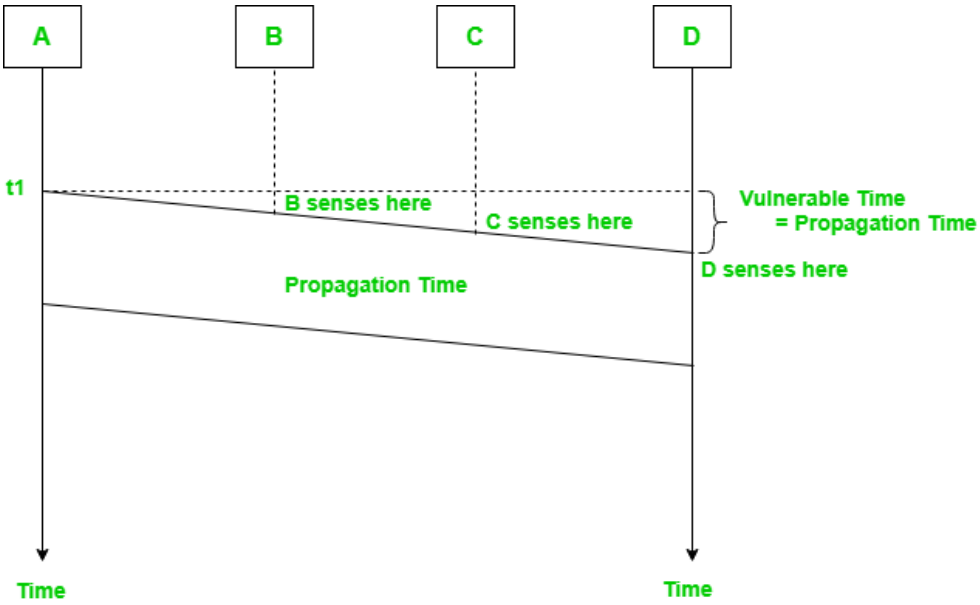
Vulnerable Time = Frame transmission time
Throughput = G exp{-*G}
Maximum throughput = 0.368 for G=1



Slotted ALOHA

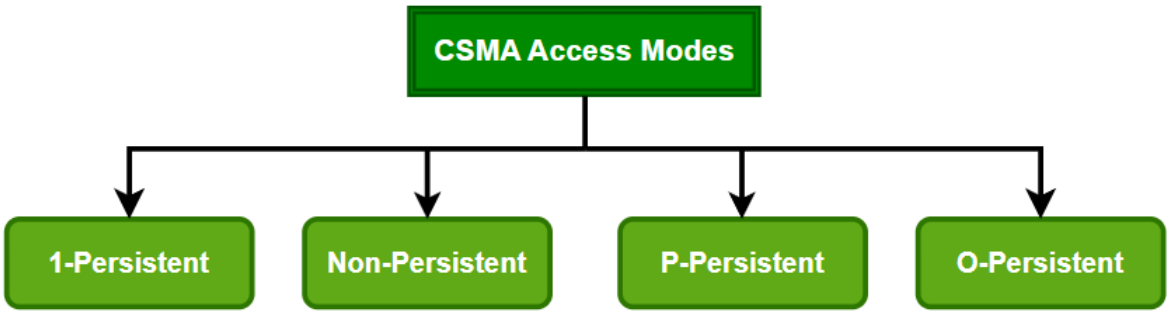
CSMA

Carrier Sense Multiple Access ensures fewer collisions as the station is required to first sense the medium (for idle or busy) before transmitting data. If it is idle then it sends data, otherwise it waits till the channel becomes idle. However there is still chance of collision in CSMA due to propagation delay. For example, if station A wants to send data, it will first sense the medium. If it finds the channel idle, it will start sending data. However, by the time the first bit of data is transmitted (delayed due to propagation delay) from station A, if station B requests to send data and senses the medium it will also find it idle and will also send data. This will result in collision of data from station A and B.



CSMA

CSMA Access Modes



CSMA Access Modes

- **1-Persistent:** The node senses the channel, if idle it sends the data, otherwise it continuously keeps on checking the medium for being idle and transmits unconditionally (with 1 probability) as soon as the channel gets idle.

- **Non-Persistent:** The node senses the channel, if idle it sends the data, otherwise it checks the medium after a random amount of time (not continuously) and transmits when found idle.
- **P-Persistent:** The node senses the medium, if idle it sends the data with p probability. If the data is not transmitted ((1-p) probability) then it waits for some time and checks the medium again, now if it is found idle then it send with p probability. This repeat continues until the frame is sent. It is used in Wifi and packet radio systems.
- **O-Persistent:** Superiority of nodes is decided beforehand and transmission occurs in that order. If the medium is idle, node waits for its time slot to send data.

CSMA/CD

Carrier sense multiple access with collision detection. Stations can terminate transmission of data if collision is detected. For more details refer – Efficiency of CSMA/CD.

CSMA/CA

Carrier sense multiple access with collision avoidance. The process of collisions detection involves sender receiving acknowledgement signals. If there is just one signal(its own) then the data is successfully sent but if there are two signals(its own and the one with which it has collided) then it means a collision has occurred. To distinguish between these two cases, collision must have a lot of impact on received signal. However it is not so in wired networks, so CSMA/CA is used in this case.

CSMA/CA Avoids Collision By

- **Interframe Space:** Station waits for medium to become idle and if found idle it does not immediately send data (to avoid collision due to propagation delay) rather it waits for a period of time called Interframe space or IFS. After this time it again checks the medium for being idle. The IFS duration depends on the priority of station.
- **Contention Window:** It is the amount of time divided into slots. If the sender is ready to send data, it chooses a random number of slots as wait time which doubles every time medium is not found idle. If the medium is found busy it does not restart the entire process, rather it restarts the timer when the channel is found idle again.
- **Acknowledgement:** The sender re-transmits the data if acknowledgement is not received before time-out.

2. Controlled Access

Controlled access protocols ensure that only one device uses the network at a time. Think of it like taking turns in a conversation so everyone can speak without talking over each other.

In this, the data is sent by that station which is approved by all other stations. For further details refer – Controlled Access Protocols.

3. Channelization

In this, the available bandwidth of the link is shared in time, frequency and code to multiple stations to access channel simultaneously.

- **Frequency Division Multiple Access (FDMA) –** The available bandwidth is divided into equal bands so that each station can be allocated its own band. Guard bands are also added so that no two bands overlap to avoid crosstalk and noise.
- **Time Division Multiple Access (TDMA) –** In this, the bandwidth is shared between multiple stations. To avoid collision time is divided into slots and stations are allotted these slots to transmit data. However there is a overhead of synchronization as each station needs to know its time slot. This is resolved by adding synchronization bits to each slot. Another issue with TDMA is propagation delay which is resolved by addition of guard bands. For more details refer – Circuit Switching
- **Code Division Multiple Access (CDMA) –** One channel carries all transmissions simultaneously. There is neither division of bandwidth nor division of time. For example, if there are many people in a room all speaking at the same time, then also perfect reception of data is possible if only two person speak the same language. Similarly, data from different stations can be transmitted simultaneously in different code languages.
- **Orthogonal Frequency Division Multiple Access (OFDMA) –** In OFDMA the available bandwidth is divided into small subcarriers in order to increase the overall performance, Now the data is transmitted through these small subcarriers. it is widely used in the 5G technology.

Advantages of OFDMA

- High data rates
- Good for multimedia traffic
- Increase in efficiency

Disadvantages OFDMA

- Complex to implement
- High peak to power ratio

- **Spatial Division Multiple Access (SDMA) –** SDMA uses multiple antennas at the transmitter and receiver to separate the signals of multiple users that are located in different spatial directions. This technique is commonly used in MIMO (Multiple-Input, Multiple-Output) wireless communication systems.

Advantages SDMA

- Frequency band uses effectively
- The overall signal quality will be improved
- The overall data rate will be increased

Disadvantages SDMA

- It is complex to implement
- It require the accurate information about the channel

Features of Multiple Access Protocols

- **Contention-Based Access:** Multiple access protocols are typically contention-based, meaning that multiple devices compete for access to the communication channel. This can lead to collisions if two or more devices transmit at the same time, which can result in data loss and decreased network performance.
- **Carrier Sense Multiple Access (CSMA):** CSMA is a widely used multiple access protocol in which devices listen for carrier signals on the communication channel before transmitting. If a carrier signal is detected, the device waits for a random amount of time before attempting to transmit to reduce the likelihood of collisions.
- **Collision Detection (CD):** CD is a feature of some multiple access protocols that allows devices to detect when a collision has occurred and take appropriate action, such as backing off and retrying the transmission.
- **Collision Avoidance (CA):** CA is a feature of some multiple access protocols that attempts to avoid collisions by assigning time slots to devices for transmission.
- **Token Passing:** Token passing is a multiple access protocol in which devices pass a special token between each other to gain access to the communication channel. Devices can only transmit data when they hold the token, which ensures that only one device can transmit at a time.
- **Bandwidth Utilization:** Multiple access protocols can affect the overall bandwidth utilization of a network. For example, contention-based protocols may result in lower bandwidth utilization due to collisions, while token passing protocols may result in higher bandwidth utilization due to the controlled access to the communication channel.

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